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Modelling Fluid-Structure Interaction using Smoothed Particle Hydrodynamics

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Title: Modelling Fluid-Structure Interaction using Smoothed Particle Hydrodynamics

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Abstract: In this thesis, we present a novel approach to the fluid-structure interaction (FSI) problems using the Symmetric Hyperbolic Thermodynamically Compatible (SHTC) equations in the Smoothed Particle Hydrodynamics (SPH) framework. Since both the fluid and the solid are described by a single system of equations the implementation and coupling became straightforward. We perform a series of numerical tests to validate the method and demonstrate its capabilities.

Keywords: Fluid-structure interaction (FSI), Smoothed Particle Hydrodynamics (SPH), Symmetric Hyperbolic thermodynamically compatible (SHTC) equations

Název práce: Modelování interakce tekutin a pevných látek pomocí Smoothed Particle Hydrodynamics

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Abstrakt: V této práci představujeme nový přístup k problémům modelování interakce tekutin a pevných látek (FSI) pomocí rovnic Symetrického Hyperbolického Termodynamicky kompatibilního (SHTC) typu v rámci metody Smoothed Particle Hydrodynamics (SPH). Jelikož jak tekutina, tak i pevná látka jsou popsány jedním systémem rovnic, implementace a coupling se stávají přímočarými. Provedli jsme řadu numerických testů pro ověření metody a demostraci jejích schopností.

Klíčová slova: Interakce kapalin a pevných látek (FSI), Smoothed Particle Hydrodynamics (SPH), Symetrické Hyperbolické Termodynamicky kompatibilní (SHTC) rovnice

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1 Introduction

In this thesis, we investigate a novel approach to fluid-structure interaction (FSI) problems using the unified model of continuum mechanics also called the Symmetric Hyperbolic Thermodynamically Compatible equations (SHTC) within the Smoothed Particle Hydrodynamics (SPH) framework. The novelty of this approach lies in the fact that both the fluid and the solid are governed by a single system of equations, in a single monolithic framework. This means that the coupling between the fluid and the solid becomes straightforward.

In this thesis, we will firstly introduce the SHTC model [1, 2, 3] which can describe flows of Newtonian and non-Newtonian fluids as well as elastoplastic solids in the first chapter.

In the second chapter, we will summarize the SPH method and the SHTC-SPH method developed by Kincl [4, 5, 6, 7]. We will also review the standard handling of the coupling between the fluid and the solid in the SPH framework.

In the third chapter, we will perform a series of simple numerical test cases for both fluid and solid deformation problems to validate the method and to demonstrate its capabilities in simulating fluid or solid deformation problems. All numerical tests are performed using the SmoothedParticles.jl framework.

Lastly, we will attempt to simulate the Turek-Hron benchmark [8] using our method and discuss the downfalls and possible improvements of our method.

1.1 SHTC

The Symmetric Hyperbolic Thermodynamically Compatible equations (SHTC), sometimes called the Unified formulation of continuum fluid and solid mechanics or the Godunov-Peskhov-Romenski (GPR) model, were first formulated in [2, 1]. The model can describe flows of Newtonian and non-Newtonian flows [9], as well as elastoplastic solids [10]. The equations were originally proposed by Godunov and Romenski in [11] and were used to model large elastoplastic deformations in metals. Similar equations and the idea to apply them to model fluid were used by Besseling in [12], however they were not appreciated in the fluid mechanics perhaps because of the dominant role of the Navier-Stokes equations.

The main difference between SHTC and other continuum models is that we assume that the continuum particles [1] have some small but finite size. The continuum particle is understood as a large enough ensemble of molecules or atoms, and the model describes the way these particles rearrange. The particles have some inner structure that allows deformations and they can exist some finite time. The lifetime of particles is an important measure of fluidity, and the deformability of particles is what allows the flow in the first place. More specifically, flow is a process of rearrangement of continuum particles [2]. Thus in SHTC the continuum is represented by a finite number of finitely small continuum particles, that cover the whole space without any gaps.

[1] Sometimes also called material elements or fluid particles

[2] Here is where the finite size of particles plays a crucial role. For infinitelly small particles, there is no notion of neighbouring particles.

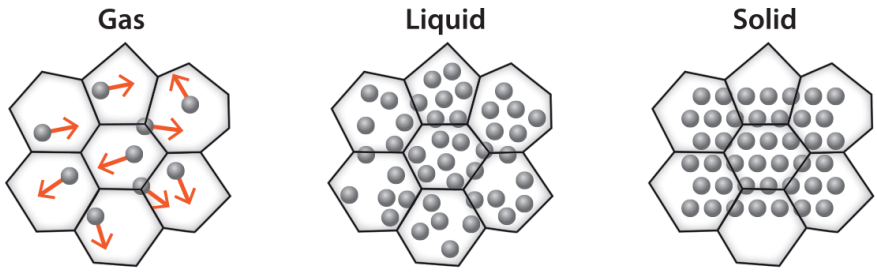


Figure 1.1 The continuum is represented by a system of particles (honeycomb-like cells), the real molecules are depicted by circles. Note that all three states of mater are described in the same way same in our representation.

To describe the deformation of continuum particles we need few ingredients. The ability of particles to deform can be characterized by the ability of particles to transmit transversal perturbations. A quantity that allows us to describe transversal deformations is the shear sound speed denoted by c_s. To describe the deformation of the continuum particles, we use the distortion field $\mathbb{A}$ that maps a particle from the current deformed configuration to the referential undeformed one.

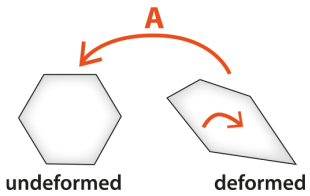


Figure 1.2 The distortion field $\mathbb{A}$ maps a particle from the current deformed configuration to the referential undeformed one.

It is important to note that the field $\mathbb{A}$ is not integrable, i.e. it does not relate Eulerian and Lagrangian coordinate frame and therefore cannot be seen as an inverse of the deformation gradient [2, 13].

The ability of the particles to rearrange, that is, to flow is characterized by time τ. [3] Time τ is a time that it takes for a continuum particle to break bonds with its neighbours. The bigger the time τ is, the more viscous is the fluid. Fluids and solids can be characterized by τ in the following way:

- For $\tau = \infty$ we have elastic solids
- For $0 < \tau < \infty$ we have viscous or viscoelastic fluids
- For $\tau \to 0$ we have ideal fluids

In summary, it is the particle representation of the continuum that allows us to describe all states of matter in a unified way. The problem of rearrangement of particles or molecules becomes a geometrical problem of deformation of the continuum particles and the deformation of a continuum particle is indenpendent of its inner structure.

[3] Inspired by Frenkel's idea of particle settled life time [14].

2 SPH

2.1 SPH

Smoothed-Particle Hydrodynamics (SPH) is a Lagrangian method that works by dividing the continuum into N particles. Each particle a has its position $\mathbf{x}_a$, velocity $\mathbf{v}_a$, mass m_a, density ρ_a etc. Given the lagrangian structure of SPH it follows that $\dot{\mathbf{x}}_a = \mathbf{v}_a$. The particles interact with each other via the *smoothing kernel* function $W_h : \mathbb{R}^d \to [0, \infty)$. The parameter $h > 0$ is called *smoothing length*. The kernel function has the following properties:

- $W_h \in C^2(\mathbb{R}^d)$
- $\int_{\mathbb{R}^d} W_h(\mathbf{x})\, d\mathbf{x} = 1$
- W_h is radial function i.e. $W_h = W_h(|\mathbf{x}|)$
- $h : W_h(r) = \frac{1}{h^d} W_1(r/h)$
- $\frac{dW_h}{dr} \leq 0$

Smoothing kernel is a discrete approximation of the Dirac delta function i.e. as $h \to 0$ we have $W_h \to \delta$ in the distributional sense.

The idea behind SPH is the following: The first step is to approximate the function f or its derivative $\partial_i f$ with respect to the i-th variable using convolution with the smoothing kernel W_h. We have the following approximation:

$$\begin{aligned} f = f * \delta \approx f * W_h, \\ \partial_i f = \partial_i f * \delta \approx \partial_i f * W_h = f * \partial_i W_h. \end{aligned} \tag{2.1}$$

The second step is to approximate the convolution using numerical quadrature with nodes $\mathbf{x}_b$. We have the following approximation:

$$\begin{aligned} (f * W_h)(\mathbf{x}_a) = \int f(\mathbf{y}) W_h(\mathbf{x}_a - \mathbf{y}) d\mathbf{y} \approx \sum_b V_b f(\mathbf{x}_b) W_h(\mathbf{x}_a - \mathbf{x}_b), \\ (f * \partial_i W_h)(\mathbf{x}_a) = \int f(\mathbf{y}) \partial_i W_h(\mathbf{x}_a - \mathbf{y}) d\mathbf{y} \approx \sum_b V_b f(\mathbf{x}_b) \partial_i W_h(\mathbf{x}_a - \mathbf{x}_b). \end{aligned} \tag{2.2}$$

Where $V_b = \rho_b / m_b$ is the volume of the b-th particle and we are summing over all particles in the system. It is convenient to introduce the following notation:

- Instead of $\mathbf{x}_a - \mathbf{x}_b$ we will write $\mathbf{x}_{ab}$
- We will write $r_{ab} = |\mathbf{x}_a - \mathbf{x}_b| = |\mathbf{x}_{ab}|$
- If there is no room for disambiguation, we will omit index h and simply write $W_{ab} = W_{h,ab}$
- We will use the notation $W'_{ab} = \frac{dW}{dr}(r_{ab})$

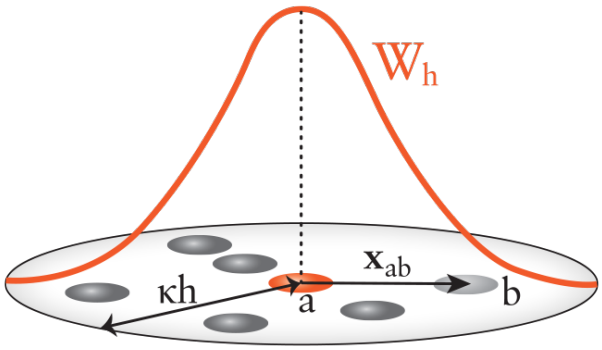


Figure 2.1 A schematic drawing of SPH kernel. κh is the size of kernel support. In case of Wendland's quintic kernel $\kappa = 2$.

For gradient of the smoothing kernel W_{ab} we have:

$$\nabla W_{ab} = W'_{ab}\frac{\mathbf{x}_{ab}}{r_{ab}} = \frac{dW}{dr}(|\mathbf{x}_{ab}|)\frac{\mathbf{x}_{ab}}{|\mathbf{x}_{ab}|} \tag{2.3}$$

A useful property of the kernel gradient is that it is anti-symmetric in indexes a and b i.e. $\nabla W_{ab} = -\nabla W_{ba}$.

As for the choice of smoothing kernel there is a wide variety of smoothing kernels one can use for example the Gaussian kernel [15],

$$W_h(r) = \frac{1}{(2\pi h^2)^{d/2}} \exp\left(-\frac{r^2}{h^2}\right) \tag{2.4}$$

the B-spline kernel, the Wendland's quintic kernel [16].

$$W_h(r) = \begin{cases} \frac{\alpha_d}{h^d}(1-\frac{r}{2h})^4(1+\frac{4r}{h}) & 0 \le r \le 2h, \\ 0 & 2h < r. \end{cases} \tag{2.5}$$

Where α_d is the normalization constant and d is the dimension of the problem.

$$\alpha_d = \begin{cases} \frac{7}{4}\pi & d = 2, \\ \frac{21}{16}\pi & d = 3 \end{cases} \tag{2.6}$$

A nice summary can be found in [17]. It is usually advantageous to pick a kernel with compact support, that way the sums in 2.2 are only over the particles that are within the support of the kernel, making the whole process computationally more efficient.

2.1.1 Weakly Compressible SPH

Weakly Compressible SPH or WCSPH for short is SPH approximation of inviscid isentropic compressible flow. The governing equations are the continuity equation and the momentum equation:

$$\begin{aligned} \frac{D\rho}{Dt} &= -\rho\nabla\cdot\mathbf{v} \\ \frac{D\mathbf{v}}{Dt} &= -\frac{1}{\rho}\nabla p \end{aligned} \tag{2.7}$$

where $\frac{D\cdot}{Dt}$ is material derivative, ρ is density, $\mathbf{v}$ is velocity in Eulerian frame and p is pressure.

By simply following the procedure outlined in the previous section and using that $\rho\nabla f = \nabla(\rho f) - f\nabla\rho$ for any two sufficiently smooth scalar functions f and ρ we could approximate the pressure gradient as follows:

$$\rho_a \nabla p_a = \sum_b (p_b - p_a) m_b \nabla W_{ab} \tag{2.8}$$

it has the advantage that the force vanishes for constant pressure however it does not conserve linear and angular momentum [18] and henceforth is seldom used.

Following [18] we can symmetrize the term by first writing:

$$\frac{\nabla p}{\rho} = \nabla\left(\frac{p}{\rho}\right) + \frac{p}{\rho^2}\nabla\rho \tag{2.9}$$

if we then use 2.2 we get the SPH momentum equation for particle a:

$$\frac{D\mathbf{v}_a}{Dt} = -\sum_b m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2}\right)\nabla W_{ab} \tag{2.10}$$

This choice is not unique in fact infinitely many symmetrizations are possible [18]. Noting that

$$\frac{\nabla p}{\rho} = \frac{p}{\rho^\sigma}\nabla\left(\frac{1}{\rho^{1-\sigma}}\right) + \frac{1}{\rho^{2-\sigma}}\nabla\left(\frac{p}{\rho^{\sigma-1}}\right) \tag{2.11}$$

We get the following version of the SPH momentum equation:

$$\frac{D\mathbf{v}_a}{Dt} = -\sum_b m_b \left(\frac{p_a}{\rho_a^\sigma \rho_b^{2-\sigma}} + \frac{p_b}{\rho_a^{2-\sigma}\rho_b^\sigma}\right)\nabla W_{ab} \tag{2.12}$$

Following the same procedure for ρ we can derive the SPH continuity equation:

$$\frac{D\rho_a}{Dt} = -\sum_b m_b \mathbf{v}_{ab}\cdot\nabla W_{ab} \tag{2.13}$$

Note that if we integrate the continuity equation we get:

$$\rho_a = \sum_b m_b W_{ab} + C_{\rho_a} \tag{2.14}$$

which is the same as if we would have used the kernel to approximate the density in the first place. The constant C_{ρ_a} is determined by the initial conditions of the problem.

The last missing piece is the equation of state. For water we can use the Tait equation of state:

$$p(\rho) = \frac{\rho_0 c^2}{\gamma}\left(\left(\frac{\rho}{\rho_0}\right)^\gamma - 1\right) + p_0 \tag{2.15}$$

where ρ_0 is the reference density, c is the speed of sound, γ is usually set to 7 and p_0 is the background pressure added to avoid negative values of pressure.

The following system of ordinary differential equations is called Weakly Compressible SPH:

$$\frac{D\rho_a}{Dt} = -\sum_b m_b \mathbf{v}_{ab}\cdot\nabla W_{ab} \tag{2.16a}$$

$$\frac{D\mathbf{v}_a}{Dt} = -\sum_b m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2}\right) \nabla W_{ab} \tag{2.16b}$$

$$p_a = \frac{\rho_0 c^2}{\gamma}\left(\left(\frac{\rho_a}{\rho_0}\right)^\gamma - 1\right) + p_0 \tag{2.16c}$$

one of the merits of using SPH in the first place are it's conservational properties.

Conservational properties

In this section we will take a closer look on WCSPH conservational properties. Conservation of linear momentum follows from antisymmetry of the kernel gradient:

$$\frac{d}{dt}\sum_a m_a \mathbf{v}_a = \sum_a m_a \frac{D\mathbf{v}_a}{Dt} = -\sum_{a,b} m_a m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2}\right) \nabla W_{ab} = 0 \tag{2.17}$$

Conservation of angular momentum follows from the fact that $\mathbf{x}_{ab}$ and $\nabla W_{ab} = W'_{ab}\frac{\mathbf{x}_{ab}}{|\mathbf{x}_{ab}|}$ are parallel:

$$\begin{aligned}\frac{d}{dt}\sum_a m_a \mathbf{x}_a \times \mathbf{v}_a = -\sum_{a,b} m_a m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2}\right) \mathbf{x}_a \times \nabla W_{ab} = \\ -\frac{1}{2}\sum_{a,b} m_a m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2}\right) \mathbf{x}_{ab} \times \nabla W_{ab} = 0\end{aligned} \tag{2.18}$$

The total energy of the system is given by:

$$E = \sum_a m_a \epsilon_a \tag{2.19}$$

where $\epsilon = \epsilon(\rho, \mathbf{v})$ is the specific energy satisfying $\frac{\partial \epsilon}{\partial \mathbf{v}} = \mathbf{v}$ and $\frac{\partial \epsilon}{\partial \rho} = \frac{p}{\rho^2}$.

Using the antisymmetry of SPH kernel it is easy to show that:

$$\begin{aligned}\frac{dE}{dt} = \frac{d}{dt}\sum_a m_a \epsilon_a = \sum_a \left(\mathbf{v}_a \cdot \frac{D\mathbf{v}}{Dt} + \frac{p_a}{\rho_a^2}\frac{D\rho_a}{Dt}\right) = \\ -\sum_{a,b} m_a m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2}\right) \mathbf{v}_a \cdot \nabla W_{ab} + \sum_{a,b} m_b m_a \frac{p_a}{\rho_a^2} \mathbf{v}_{ab} \cdot \nabla W_{ab} = 0\end{aligned} \tag{2.20}$$

Variational principle

Given the Lagrangian nature of SPH we can derive the equations of motion from the principle of least action, starting with the following Lagrangian

$$\mathcal{L} = \sum_b m_b \left(\frac{1}{2}\mathbf{v}_b^2 - e_b\right) \tag{2.21}$$

where $e = e(\rho)$ is the specific internal energy satisfying $\frac{\partial e}{\partial \rho} = \frac{p}{\rho^2}$. The Euler-Lagrange equations for each particle a than read as:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \mathbf{v}_a} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}_a} \tag{2.22}$$

Plugging $\mathcal{L}$ into the equation above we get:

$$m_a \frac{d\mathbf{v}_a}{dt} = -\sum_b m_b \frac{\partial e_b}{\partial \mathbf{x}_a} = -\sum_b m_b \frac{\partial e_b}{\partial \rho_b}\frac{\partial \rho_b}{\partial \mathbf{x}_a} = -\sum_b m_b \frac{p_b}{\rho_b^2}\frac{\partial \rho_b}{\partial \mathbf{x}_a} \tag{2.23}$$

Using SPH interpolation of density ρ_b we get:

$$\frac{\partial \rho_b}{\partial \mathbf{x}_a} = \frac{\partial}{\partial \mathbf{x}_a}\sum_c m_c W_{bc} = \begin{cases} \sum_c m_c \nabla W_{bc} & b = a, \\ -m_a \nabla W_{ab} & b \neq a \end{cases} \tag{2.24}$$

Proceeding with the calculation we get:

$$\begin{aligned} m_a \frac{d\mathbf{v}_a}{dt} = -\sum_b m_b \frac{p_b}{\rho_b^2}\frac{\partial \rho_b}{\partial \mathbf{x}_a} = -m_a \frac{p_a}{\rho_a^2}\sum_c m_c \nabla W_{ac} - \sum_{b\neq a} m_b \frac{p_b}{\rho_b^2} m_a \nabla W_{ab} = \\ -\sum_b m_a m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2}\right)\nabla W_{ab} \end{aligned} \tag{2.25}$$

which is the same as the SPH momentum equation 2.10.

The conservation of linear and angular momentum then follows from the invariance of the Lagrangian $\mathcal{L}$ to translations and rotations.

Viscous flow

So far we have discussed inviscid flow. To obtain Navier-Stokes equations we need to add a viscous term $\mathbf{\Pi}$ of the form:

$$\mathbf{\Pi} = \mu \nabla \cdot \left(\nabla \mathbf{v} + (\nabla \mathbf{v})^T\right) \tag{2.26}$$

to the momentum equation 2.7, where μ is the shear viscosity. One could try to use the second derivative of the smoothing kernel, however it is not recommended [17] as second derivative of the smoothing kernel is usually ill behaved. There are several available workarounds.

One is the artificial viscosity term developed by Gingold and Monaghan [15] that was originally suggested to enhance the stability of the SPH method. The idea is to modify the momentum equation 2.10 as:

$$\frac{D\mathbf{v}_a}{Dt} = -\sum_b m_b \left(\frac{p_a}{\rho_a^2} + \frac{p_b}{\rho_b^2}\right)\nabla W_{ab} + \mathbf{\Pi}_a \tag{2.27}$$

where $\mathbf{\Pi}_a$ is the artificial viscosity term given by:

$$\mathbf{\Pi}_a = -2\alpha h c \sum_b \frac{m_b}{\rho_a + \rho_b}\frac{\mathbf{v}_{ab}\cdot \mathbf{x}_{ab}}{r_{ab}^2 + \xi^2}\nabla W_{ab} \tag{2.28}$$

where α is artificial viscosity coefficient, c is the speed of sound, ξ is a small number to avoid division by zero. It was shown by Colagrossi et al. [19] that when artificial viscosity is used it is equivalent to having a Newtonian viscous term in the momentum equation. An SPH equivalent of Reynolds number in 2D can be then estimated as:

$$\mathrm{Re} = \frac{\rho U_{max} L}{\mu} \approx \frac{8 U_{max} L}{\alpha h c} \tag{2.29}$$

where U_{max} is the maximum velocity in the system and L is the characteristic length of the system.

Another choice for the viscosity term is:

$$\mathbf{\Pi}_a = -\sum_b \frac{m_b}{\rho_a \rho_b}(\mu_a + \mu_b)\mathbf{v}_{ab}\frac{\mathbf{x}_{ab}\cdot\nabla W_{ab}}{r_{ab}^2} \tag{2.30}$$

firstly proposed by [20]. This term models incompressible viscous flow however, it does not conserve angular momentum and is therefore usually not preferred.

2.1.2 Fluid-structure interaction

Traditionally SPH is used for fluid dynamics, and when it comes to fluid-structure interaction (FSI) SPH is usually coupled with another method. See, for example, [21] for SPH-DEM, [22] for SPH-FEM or [23] for SPH-FVM.

Simulating FSI entirely within SPH was proposed by Antoci et al. [24]. The idea was then refined by Hu et al. in [25, 26, 27] and we will present it here.

Fluid-structure interaction

We assume the fluid to be isothermal, incompressible, and viscous. It is assumed that the solid structure is weakly-compressible and elastic. However, we are mostly interested in the coupling between the fluid and the solid structure and the evolution equations for the solid structure are not of great importance to us. For more details on handling the solid structure, we refer to [26, 27].

Including FSI forces the momentum equation for fluid can be written as:

$$\frac{D\mathbf{v}_a}{Dt} = \mathbf{f}_a^{f:p} + \mathbf{f}_a^{f:\nu} + \mathbf{f}_a^{s:p} + \mathbf{f}_a^{s:\nu} \tag{2.31}$$

where $\mathbf{f}_a^f$ is the unit force on fluid particle a due to other fluid particles, $\mathbf{f}_a^s$ is the unit force on fluid particle a due to structure particles. Both forces are divided into pressure and viscous part i.e. $\mathbf{f}_a^f = \mathbf{f}_a^{f:p} + \mathbf{f}_a^{f:\nu}$ and $\mathbf{f}_a^s = \mathbf{f}_a^{s:p} + \mathbf{f}_a^{s:\nu}$.

Similarly the momentum equation for structure particles can be written as:

$$\frac{D\mathbf{v}_b}{Dt} = \mathbf{f}_b^s + \mathbf{f}_b^{f:p} + \mathbf{f}_b^{f:\nu} \tag{2.32}$$

where $\mathbf{f}_b^s$ is the unit force on structure particle b due to other structure particles that is determined by the solid material model and $\mathbf{f}_b^{f:p}$ and $\mathbf{f}_b^{f:\nu}$ are the forces on structure particle a due to fluid particles.

For any fluid particle a and solid particle b we can write the forces acting on fluid particle a due to solid particle b by using Newton's third law as:

$$\begin{aligned} \mathbf{f}_a^{b:p} &= -\mathbf{f}_b^{a:p} \\ \mathbf{f}_a^{b:\nu} &= -\mathbf{f}_b^{a:\nu} \end{aligned} \tag{2.33}$$

For the proper resolution of the FSI forces it is enough to evaluate $\mathbf{f}_a^{s:p}$ and $\mathbf{f}_a^{s:\nu}$ for the fluid particles and we get $\mathbf{f}_b^{f:p}$ and $\mathbf{f}_b^{f:\nu}$ for free from the Newton's third law.

The boundary conditions and FSI forces are modeled by the so-called *dummy particles* which are placed in several layers and used as walls. The dummy particles

follow the same equations of motion as the fluid particles but are not allowed to move. Other approaches exist however, they are out-shined by the dummy particle method in terms of simplicity and efficiency when implementing complex geometries.

We will assume that the dummy particles used to model the fluid-structural forces $\mathbf{f}^s = \mathbf{f}^{s:p} + \mathbf{f}^{s:\nu}$ coincide with the structure particles. We will denote the internal parameters of the dummy particle with index d that coincides with the structure particle b as $m_b^d = m_b$, $\mathbf{x}_b^d = \mathbf{x}_b$, $\mathbf{v}_b^d$, p_b^d, etc.

For WCSPH we can write the pressure part of the force $\mathbf{f}_a^{s:p}$ as:

$$\mathbf{f}_a^{s:p} = -\sum_b m_b^d \left(\frac{p_a}{\rho_a^2} + \frac{p_b^d}{(\rho_b^d)^2} \right) \nabla W_{ab} \tag{2.34}$$

and the viscous part of the force $\mathbf{f}_a^{s:\nu}$ as:

$$\mathbf{f}_a^{s:\nu} = -2\mu \sum_b \frac{m_a m_b^d}{\rho_a \rho_b^d} \frac{1}{r_{ab}} \frac{\partial W_{ab}}{\partial r_{ab}} (\mathbf{v}_a - \mathbf{v}_b^d) \tag{2.35}$$

where index d runs over all dummy particle, p_b^d is the imaginary pressure of dummy particle d and $\mathbf{v}_b^d$ is the imaginary velocity of dummy particle d.

The imaginary parameters of the dummy particle d p_b^d and $\mathbf{v}_b^d$ have nothing to do with the actual pressure and velocity of the structure particle b that coincides with the dummy particle d. They play no part in the evolution of the motion of the solid structure part and are used only to model the FSI forces and boundary conditions.

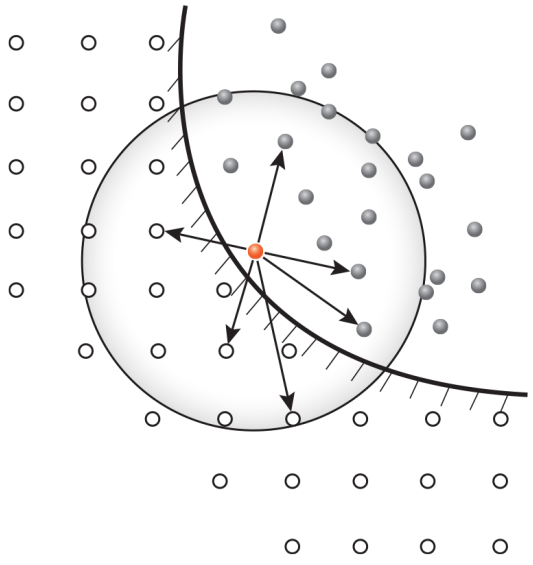


Figure 2.2 Fluid particles • interact with dummy particles ○ representing the walls. The dummy particles are placed in several layers to prevent kernel cutoff.

Boundary conditions

So far we have not specified the boundary conditions at the wall. The free-slip or no-slip boundary conditions can be enforced by the choice of $\mathbf{v}_b^d$ in the viscous

part of the force. By omiting the viscous interaction of a fluid particle with the surrounding dummy particles we can enforce free-slip boundary condition.

If we want to enforce no-slip boundary condition we can use the following approximation of $\mathbf{v}_b^d$ [25]:

$$\mathbf{v}_b^d = 2\mathbf{v}_b - \tilde{\mathbf{v}}_a \tag{2.36}$$

where $\tilde{\mathbf{v}}_a = \sum_c \mathbf{v}_c W_{ac}$. Another choice for $\mathbf{v}_b^d$ suggested in [7] is:

$$\mathbf{v}_b^d = 2\mathbf{v}_b - \mathbf{v}_a \tag{2.37}$$

Recall that the no-slip boundary condition means that a viscous fluid has zero bulk velocity at the boundary. That is, the velocity of the fluid at the boundary matches the velocity of the boundary itself.

For a structure particle b with velocity $\mathbf{v}_b$ we can approximate the no-slip boundary condition as:

$$\mathbf{v}_{interface} \approx \mathbf{v}_b = \frac{1}{2}\left(\mathbf{v}_b^d + \mathbf{v}_a\right) \tag{2.38}$$

or:

$$\mathbf{v}_{interface} \approx \mathbf{v}_b = \frac{1}{2}\left(\mathbf{v}_b^d + \tilde{\mathbf{v}}_a\right) \tag{2.39}$$

which is consistent with our choice of $\mathbf{v}_b^d$.

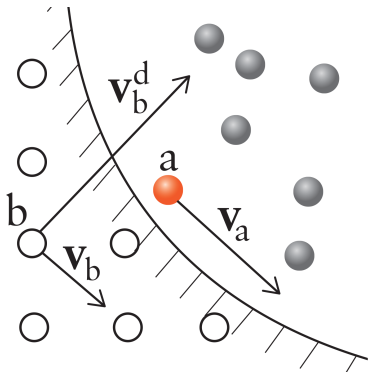


Figure 2.3 Fluid particles a interacts with the solid particle b that coincides with the dummy particle d. Velocity $\mathbf{v}_a$ is the velocity of the fluid particle, $\mathbf{v}_b^d$ is the velocity of the dummy particle d that ensures the boundary conditions and prevents penetration of the wall. $\mathbf{v}_b$ is the actual velocity of the solid particle b.

Assuming no-slip boundary condition we can take the imaginary values as follows [7]:

$$\begin{cases} p_b^d = p_a \\ \mathbf{v}_b^d = 2\mathbf{v}_b - \mathbf{v}_a \end{cases} \tag{2.40}$$

Further refinement has been done in [26] based on the idea in [25]:

$$\begin{cases} p_b^d = p_a + \max(0, (\mathbf{g} - \frac{D\mathbf{v}_a}{Dt}) \cdot \mathbf{n}^s)\,(\mathbf{x}_{ab} \cdot \mathbf{n}^s) \\ \mathbf{v}_b^d = 2\mathbf{v}_b - \mathbf{v}_a \end{cases} \tag{2.41}$$

where $\mathbf{n}^s$ is the normal vector of the structure, $\mathbf{g}$ is the body force e.g. gravity and the imaginary density ρ_b^d in 2.34 and 2.35 is recovered from equation of state 2.15.

In this thesis, we opt for a different and completely new approach. Even though our aproach might not be that well suited for simulating FSI problems where the difference between the solid and fluid is clear cut, it might prove advantageous when simulating materials that are acting as both fluid and solid and where the approach mentioned above might not be that well suited. We will discuss this aproach in the following section.

2.2 SHTC-SPH

2.2.1 SHTC-SPH method

The SHTC-SPH method developed by Kincl et al. in [5] aims to find approximative solutions of the SHTC equations in both solid and fluid regimes using SPH machinery. In this section, we will present the method.

Recall that SHTC equations sometimes called the Unified model of continuum and solid mechanics [2, 1] are formulated as follows:

$$\frac{D\rho}{Dt} + \rho\nabla\cdot\mathbf{v} = 0 \tag{2.42a}$$

$$\rho\frac{D\mathbf{v}}{Dt} - \nabla\cdot\mathbb{S} = 0 \tag{2.42b}$$

$$\frac{D\mathbb{A}}{Dt} + \mathbb{A}\nabla\mathbf{v} = -\frac{1}{\theta}E_{\mathbb{A}} \tag{2.42c}$$

$$\rho\frac{DE}{Dt} - \nabla\cdot(\mathbb{S}\mathbf{v}) = 0 \tag{2.42d}$$

where $\mathbb{S}$ is the total stress tensor that depends on the material under consideration, $\mathbb{A}$ is the distortion field, $\theta = \theta(\rho, \tau, \mathbb{A}) \geq 0$ is the relaxation function and $E = E(\rho, \mathbf{v}, \mathbb{A}) = \epsilon(\rho, \mathbb{A}) + \frac{1}{2}\|\mathbf{v}\|^2$ is the total energy density and $\epsilon(\rho, \mathbb{A})$ is the internal energy that depends on the material under consideration.

Discrete distortion

Distortion field $\mathbb{A}$ plays a crucial role in the SHTC equations. In case 2.42c is homogenous i.e. the process is reversible (elastic), the distortion field can be though of as the inverse of the deformation gradient $\mathbb{F} = \frac{\partial\mathbf{x}}{\partial\mathbf{X}} = \mathbb{A}^{-1}$, where $\mathbf{X}$ is the Lagrangian coordinate of continuum and it satisfies the following equality:

$$\frac{D\mathbb{A}}{Dt} = -\mathbb{A}\mathbb{L} \tag{2.43}$$

where $\mathbb{L}$ is the velocity gradient $\nabla\mathbf{v}$. Indeed, we can write:

$$\begin{aligned}\frac{D\mathbb{F}}{Dt} = \frac{D}{Dt}\left(\frac{\partial\mathbf{x}}{\partial\mathbf{X}}\right) = \frac{\partial}{\partial t}\left(\frac{\partial\mathbf{x}}{\partial\mathbf{X}}\right)\Big|_{\mathbf{x}} = \frac{\partial}{\partial\mathbf{X}}\left(\frac{\partial\mathbf{x}}{\partial t}\right)\Big|_{\mathbf{x}} = \\ \frac{\partial\mathbf{V}(\mathbf{X},t)}{\partial\mathbf{X}} = \frac{\partial\mathbf{v}(\mathbf{x}(\mathbf{X}),t)}{\partial\mathbf{X}} = \frac{\partial\mathbf{v}}{\partial\mathbf{x}}\frac{\partial\mathbf{x}}{\partial\mathbf{X}} = \nabla\mathbf{v}\cdot\mathbb{F} = \mathbb{L}\mathbb{F}\end{aligned} \tag{2.44}$$

where we used the fact that $\frac{D}{Dt}$ and $\frac{\partial}{\partial \mathbf{X}}$ commute for fixed $\mathbf{X}$ and we have used the Eulerian and Lagrangian velocity fields $\mathbf{v}$ and $\mathbf{V}$. Now using $\mathbb{F}\mathbb{F}^{-1} = \mathbb{I}$ and $\frac{D(\mathbb{F}\mathbb{F}^{-1})}{Dt} = \frac{D\mathbb{F}}{Dt}\mathbb{F}^{-1} + \mathbb{F}\frac{D\mathbb{F}^{-1}}{Dt} = \frac{D\mathbb{I}}{Dt} = 0$ we get:

$$\frac{D\mathbb{F}^{-1}}{Dt} = -\mathbb{F}^{-1}\mathbb{L} \tag{2.45}$$

which is the equation for the distortion field in the elastic case.

The velocity gradient $\mathbb{L}$ can be approximated as follows [5]:

$$\mathbb{L}_a = \left(\sum_b m_b \mathbf{v}_{ab} \otimes \nabla W_{ab}\right)\left(\sum_b m_b \mathbf{x}_{ab} \otimes \nabla W_{ab}\right)^{-1} \tag{2.46}$$

Evolution equation for $\mathbb{A}_a$ than takes the form:

$$\frac{D\mathbb{A}_a}{Dt} = -\mathbb{A}_a\left(\sum_b m_b \mathbf{v}_{ab} \otimes \nabla W_{ab}\right)\left(\sum_b m_b \mathbf{x}_{ab} \otimes \nabla W_{ab}\right)^{-1} \tag{2.47}$$

Momentum equation

Consider an elastic solid with internal energy given by:

$$\mathcal{U} = \int_\Omega \rho\epsilon \mathrm{d}\mathbf{x} \tag{2.48}$$

where $\epsilon = \epsilon(\rho, \mathbb{A})$ is the specific internal energy. The SPH approximation is equal to:

$$\mathcal{U}_h = \sum_a m_a \epsilon_a \tag{2.49}$$

where $\epsilon_a = \epsilon(\rho_a, \mathbb{A}_a)$ is the specific internal energy of particle a. We want to find the differential of $\mathcal{U}_h$ with respect to position $\mathbf{x}$:

$$\mathrm{d}\mathcal{U}_h = \sum_a m_a \epsilon_{\rho_a} \mathrm{d}\rho_a + \sum_a m_a \epsilon_{\mathbb{A}_a} : \mathrm{d}\mathbb{A}_a \tag{2.50}$$

where $\epsilon_{\rho_a} = \frac{\partial \epsilon}{\partial \rho_a}$ and $\epsilon_{\mathbb{A}_a} = \frac{\partial \epsilon}{\partial \mathbb{A}_a}$ is the usual notation for partial derivatives. The total differential of density ρ_a with respect to position is given by:

$$\mathrm{d}\rho_a = \sum_b m_b \mathrm{d}W_{ab} = \sum_b m_b \sum_b \frac{\mathrm{d}W_{ab}}{\mathrm{d}r}\frac{\mathrm{d}r}{\mathrm{d}\mathbf{x}}\mathrm{d}\mathbf{x}_{ab} = \sum_b m_b \nabla W_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \tag{2.51}$$

The total differential of $\mathbb{A}_a$ with respect to position can be found using 2.47:

$$\mathrm{d}\mathbb{A}_a = -\mathbb{A}_a\left(\sum_b m_b \mathrm{d}\mathbf{x}_{ab} \otimes \nabla W_{ab}\right)\left(\sum_b m_b \mathbf{x}_{ab} \otimes \nabla W_{ab}\right)^{-1} \tag{2.52}$$

where we have used $\frac{D\mathbf{x}_a}{Dt} = \mathbf{v}_a$ and linearity of 2.47. Denoting for simplicity:

$$\begin{aligned} \mathbb{H}_a &= \sum_b m_b \mathbf{x}_{ab} \otimes \nabla W_{ab} \\ \mathbb{T}_a &= -\epsilon_{\rho_a}\mathbb{I} + \mathbb{A}_a^T \epsilon_{\mathbb{A}_a} \mathbb{H}_a^{-1} \end{aligned} \tag{2.53}$$

we can write:

$$\begin{gathered}
\mathrm{d}\mathcal{U}_h = \sum_a m_a \epsilon_{\rho_a} \mathrm{d}\rho_a + \sum_a m_a \epsilon_{\mathbb{A}_a} : \mathrm{d}\mathbb{A}_a = \\
\sum_{a,b} \left(\epsilon_{\mathbb{A}_a} \nabla W_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} - \epsilon_{\mathbb{A}_a} : \mathbb{A}_a (\mathrm{d}\mathbf{x}_{ab} \otimes \nabla W_{ab}) \mathbb{H}_a^{-1} \right) = \\
- \sum_{a,b} m_a m_b \left(-\epsilon_{\rho_a} \mathbb{I} + \mathbb{A}_a^T \epsilon_{\mathbb{A}_a} \mathbb{H}_a^{-1} \right) \nabla W_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} = \\
- \sum_{a,b} m_a m_b \mathbb{T}_a \nabla W_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} = \\
- \sum_{a,b} m_a m_b \left(\mathbb{T}_a + \mathbb{T}_b \right) \nabla W_{ab} \cdot \mathrm{d}\mathbf{x}_a
\end{gathered} \tag{2.54}$$

Using $m_a \frac{D\mathbf{v}_a}{Dt} = -\frac{\partial \mathcal{U}_h}{\partial \mathbf{x}_a}$ we get the following equation:

$$\frac{D\mathbf{v}_a}{Dt} = \sum_b m_b \left(\mathbb{T}_a + \mathbb{T}_b \right) \nabla W_{ab} \tag{2.55}$$

which is SPH equivalent of 2.42b. Matrix $\mathbb{T}_a$ represents the stress on particle a. The SHTC-SPH equations for elastic solid then read as:

$$\frac{D\rho_a}{Dt} = -\sum_b m_b \mathbf{v}_{ab} \cdot \nabla W_{ab} \tag{2.56a}$$

$$\frac{D\mathbf{v}_a}{Dt} = \sum_b m_b \left(\mathbb{T}_a + \mathbb{T}_b \right) \nabla W_{ab} \tag{2.56b}$$

$$\frac{D\mathbb{A}}{Dt} + \mathbb{A}_a \mathbb{L}_a = 0 \tag{2.56c}$$

Conservational properties

In this section we will show that SHTC-SPH conserves the same quantities as WCSPH that is linear and angular momentum and energy. Conservation of linear momentum follows from antisymmetry of W_{ab}:

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_a m_a \mathbf{v}_a = \sum_a m_a \frac{D\mathbf{v}_a}{Dt} = \sum_{a,b} m_a m_b \left(\mathbb{T}_a + \mathbb{T}_b \right) \nabla W_{ab} = 0 \tag{2.57}$$

Conservation of angular momentum follows from the symmetry of $\mathbb{T}_a \mathbb{H}_a$:

$$\begin{gathered}
\frac{\mathrm{d}}{\mathrm{d}t} \left(\sum_a m_a \mathbf{x}_a \times \mathbf{v}_a \right)^i = \epsilon_{ijk} \sum_a x_a^j v_a^k = \epsilon_{ijk} \sum_a m_a x_a^j \left(\sum_b m_b (T_a^{kl} + T_b^{kl}) \nabla W_{ab}^l \right) = \\
\epsilon_{ijk} \sum_a m_a T_a^{kl} \left(\sum_b m_b x_{ab}^j \nabla W_{ab}^l \right) = \epsilon_{ijk} \sum_a m_a T_a^{kl} H_a^{lj} = 0
\end{gathered} \tag{2.58}$$

assuming that $\mathbb{A}_a^T \epsilon_{\mathbb{A}_a}$ is symmetric. Conservation of energy follows from construction:

$$\begin{gathered}
\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \sum_a m_a \left(\frac{1}{2} \|\mathbf{v}_a\|^2 + \epsilon_a \right) = \sum_a m_a \mathbf{v}_a \cdot \frac{\mathrm{d}\mathbf{v}_a}{\mathrm{d}t} + \frac{\mathrm{d}\mathcal{U}_h}{\mathrm{d}t} = \\
- \sum_a \mathbf{v}_a \cdot \frac{\mathrm{d}\mathcal{U}_h}{\mathrm{d}\mathbf{x}_a} + \sum_a \frac{\mathrm{d}\mathcal{U}_h}{\mathrm{d}\mathbf{x}_a} \cdot \frac{\mathrm{d}\mathbf{x}_a}{\mathrm{d}t} = 0
\end{gathered} \tag{2.59}$$

Constitutive relations

For the system of equations 2.66 to be complete one has to specify the internal energy $\epsilon(\rho, \mathbb{A})$. We will asume that it can be decomposed as:

$$\epsilon(\rho, \mathbb{A}) = \epsilon_0(\rho) + \epsilon_s(\rho, \mathbb{A}) \tag{2.60}$$

where $\epsilon_0(\rho)$ is the part of internal energy that depends only on volumetric changes and $\epsilon_s(\rho, \mathbb{A})$ is the elastic part of internal energy.

It is convenient to choose ϵ_0 in such a way that a well known equation of state is recovered, when ϵ_s disappears. We use the following formula:

$$\epsilon_0(\rho) = \frac{c_0^2}{2}\left(\frac{\rho_0}{\rho} - 1\right)^2 \tag{2.61}$$

where ρ_0 is the reference density and c_0 is the bulk speed of sound in the material. The pressure will be then equal to:

$$p(\rho) = \rho^2 \frac{\partial \epsilon_0}{\partial \rho} = c_0^2 \rho_0 \left(1 - \frac{\rho_0}{\rho}\right) \tag{2.62}$$

Similarly when choosing ϵ_s one should recover some well know material model for small deformations. Following [1] we use:

$$\epsilon_s(\mathbb{A}) = \frac{c_s^2}{4}\|\mathrm{dev}(\mathbb{A}^T\mathbb{A})\|_F^2 \tag{2.63}$$

where c_s is the shear sound speed, $\|\cdot\|_F$ is the Frobenius norm and

$$\mathrm{dev}\mathbb{T} = \mathbb{T} - \frac{1}{3}(\mathrm{tr}\mathbb{T})\mathbb{I} \tag{2.64}$$

The Hook law of linear elasticity is recovered in case of small deformations. Using 2.53 the stress tensor $\mathbb{T}_a$ than takes the form:

$$\mathbb{T}_a = -\frac{p_a}{\rho_a^2}\mathbb{I} + c_s^2 \mathbb{A}_a^T \mathbb{A}_a \mathrm{dev}(\mathbb{A}_a^T \mathbb{A}_a)\mathbb{H}_a^{-1} \tag{2.65}$$

Relaxation of $\mathbb{A}$

So far, we have discussed the reversible[1] part of SHTC equations. To add fluidity to our model, there is one last step we have to do. We need to add the irreversible part of SHTC equations by relaxing $\mathbb{A}$, hence adding tangential stresses. The ODE system 2.56 then becomes:

$$\frac{D\rho_a}{Dt} = -\sum_b m_b \mathbf{v}_{ab} \cdot \nabla W_{ab} \tag{2.66a}$$

$$\frac{D\mathbf{v}_a}{Dt} = \sum_b m_b \left(\mathbb{T}_a + \mathbb{T}_b\right) \nabla W_{ab} \tag{2.66b}$$

$$\frac{D\mathbb{A}}{Dt} + \mathbb{A}_a \mathbb{L}_a = -\frac{3}{\tau c_s^2}\epsilon_{\mathbb{A}_a} \tag{2.66c}$$

The relaxation function θ in 2.42c is taken as $\theta = \frac{\tau c_s^2}{3}(\det \mathbb{A})^{-5/3}$ [1]. The factor $(\det \mathbb{A})^{-5/3}$ is important only in case of compressible viscous flows, which we will not consider, for incompressible flows it is equal to one.

[1]When we say an equation is reversible we mean it has odd parity with respect to time reversal transformation (TRT). I.e. the equation changes sign when we reverse the flow of time. For more details see [28, 29].

Time integrator

Since ODE system 2.56 conserves energy we would like to choose a time integrator that has good conservational properties. One such choice is the symplectic Verlet scheme [30]. Unfortunately the equation 2.66c is not symplectic even if there is no relaxation. If there is relaxation the equation is often stiff and therefore an implicit or exponential time integrator is recommended [1]. We will follow [5] and use a splitting strategy - Verlet for the reversible part and mid-point rule combined with Runge-Kutta-4 scheme [30] for distortion $\mathbb{A}_a$.

First we will add an evolution equation for position $\mathbf{x}_a$ to system 2.66 and rewrite it as:

$$\frac{D\mathbf{x}_a}{Dt} = \mathbf{v}_a \tag{2.67a}$$

$$\frac{D\mathbf{v}_a}{Dt} = \frac{\mathbf{f}_a}{m_a} \tag{2.67b}$$

$$\frac{D\mathbb{A}_a}{Dt} = -\mathbb{A}_a\mathbb{L}_a - \frac{3}{\tau c_s^2}\epsilon_{\mathbb{A}_a} \tag{2.67c}$$

where the force $\mathbf{f}_a$ is equal to:

$$\mathbf{f}_a = \sum_b m_a m_b \left(\mathbb{T}_a(\mathbf{x}_a, \rho_a, \mathbb{A}_a) + \mathbb{T}_b(\mathbf{x}_b, \rho_b, \mathbb{A}_b)\right) \nabla W_{ab} \tag{2.68}$$

Denoting $t_k = k\Delta t$ the k-th time-step and using the notation [2] $\mathbf{f}_a(t) = \mathbf{f}_a(\mathbf{x}(t), \mathbb{A})$ and $\mathbb{L}_a(t) = \mathbb{L}_a(\mathbf{x}(t), \mathbf{v}(t))$ the scheme can be written as:

$$\mathbf{v}_a(t_{k+1/2}) = \mathbf{v}_a(t_k) + \frac{\Delta t}{2m_a}\mathbf{f}_a(t_k) \tag{2.69a}$$

$$\mathbf{x}_a(t_{k+1/2}) = \mathbf{x}_a(t_k) + \frac{\Delta t}{2}\mathbf{v}_a(t_{k+1/2}) \tag{2.69b}$$

$$\mathbb{A}_a(t_k^*) = \mathbb{A}_a(t_k)\left(\mathbb{I} - \frac{\Delta t}{2}\mathbb{L}_a(t_{k+1/2})\right)\left(\mathbb{I} + \frac{\Delta t}{2}\mathbb{L}_a(t_{k+1/2})\right)^{-1} \tag{2.69c}$$

$$\mathbb{A}_a(t_k) = \mathbb{A}_a(t_k^*) + \frac{\Delta t}{6}\left(\mathbb{K}_{1,a}(t_k^*) + 2\mathbb{K}_{2,a}(t_k^*) + 2\mathbb{K}_{3,a}(t_k^*) + \mathbb{K}_{4,a}(t_k^*)\right) \tag{2.69d}$$

$$\mathbf{x}_a(t_{k+1}) = \mathbf{x}_a(t_{k+1/2}) + \frac{\Delta t}{2}\mathbf{v}_a(t_{k+1/2}) \tag{2.69e}$$

$$\mathbf{v}_a(t_{k+1}) = \mathbf{v}_a(t_{k+1/2}) + \frac{\Delta t}{2m_a}\mathbf{f}_a(t_{k+1}) \tag{2.69f}$$

where

$$\mathbb{K}_{i,a}(t_k^*) = -\frac{3}{\tau c_s^2}\epsilon_{\mathbb{A}_a}\Big|_{\mathbb{A}=\mathbb{A}(t_k^*)+b_i\Delta t\mathbb{K}_{i-1,a}(t_k^*)} \qquad \text{i} = 1,\ 2,\ 3,\ 4, \tag{2.70}$$

and $(b_1, b_2, b_3, b_4) = (1, 1/2, 1/2, 1)$. In case of elastic solid there is no relaxation (we have $\tau \to \infty$) and this scheme becomes reversible in time. Indeed, inverting the signs of $\mathbf{v}_a$ and $\mathbb{L}_a$ leads to the same scheme with swapped roles of t_k and t_{k+1} [6].

[2] Using the fact that ρ is a function of $\mathbf{x}$.

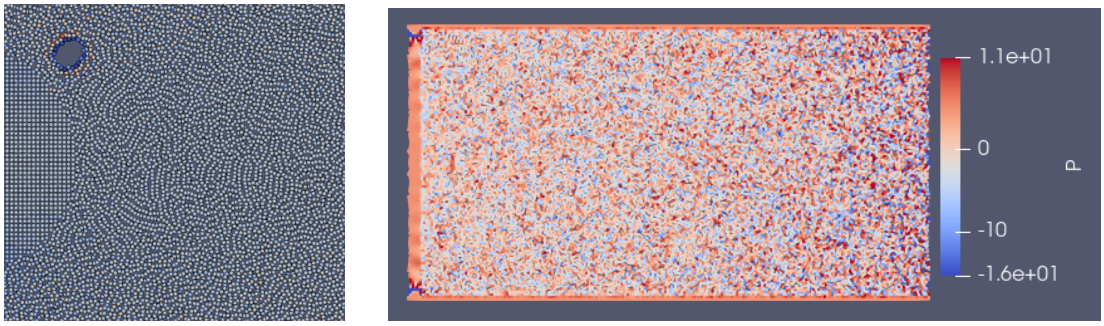
It may be worth mentioning that in case of a simulation where the particles leave the domain 'relatively quickly' it might be advantageous to switch to fully a implicit scheme as it would allow much larger time steps and the accumulated error in energy could be reduced by the short life expectancy of the particles. However, it remains to be investigated.

Tensile instability and other numerical artifacts

Tensile instability is common numerical artifact in SPH. It can lead to particle clustering and unphysical behaviour. It was shown in [31] that tensile instability is independent of the numerical time integration scheme. Under tensile strain the density ρ of the particles might increase and for negative pressures $p < 0$ the particles can reduce their potential according to

$$\frac{\partial \epsilon}{\partial \rho} = \frac{p}{\rho^2} \tag{2.71}$$

this than leads to unphysical clumping. The particle clumping is therefore noticeable mainly in materials or fluids that have state equation, that allows negative pressures. For more detailed explanation, see [31].



(a) Tensile instability **(b)** Noise in the pressure field

Figure 2.4 Tensile instability manifests itself through forming of the particle chains and cavities. The pressure field is particularly noisy near the outflow, which is to be expected.

There are multiple ways to deal with tensile instability, some of them are:

- Adding background pressure term to keep the pressure positive, effective especially in state equations that allow negative pressures Morris et al. [20].
- Monaghan [32] suggests to add a repulsive force to the particles that are to close.
- Adaptive kernel Lahiri et al. [33] that changes shape depending on stress.
- Tensile penalty Kincl. et al. [5] adding an artificial term to the energy that creates a repulsive force between the particles with the advantage that the perturbed energy is conserved.
- Transport-velocity formulation [34, 35] an algorithm that adds an additional corrective term to the velocity field.

- Tensile instability control algorithm [36] that switches between conservative and nonconservative formulation of the SPH in instability regions.
- Particle shifting techniques, or δ-SPH [37, 38]
- Particles splitting [39] could be used to fill the vacant space.

We will follow Kincl et al. [5] and use the tensile penalty method. The following *tensile penalty* term is added to the energy:

$$\mathcal{P}_h = \frac{1}{2}\sum_a m_a c_p^2 \left(\frac{\lambda_a}{\rho_0}\right)^2 \tag{2.72}$$

where c_p has a dimension of velocity and determines the strength of the anticlumping forces. The variable λ_a is defined as:

$$\lambda_a = h\frac{\partial \rho_a}{\partial h} + C_{\lambda_a} = h\sum_b m_b \frac{\partial W_{ab}}{\partial h} + C_{\lambda_a} \tag{2.73}$$

and C_{λ_a} ensures that $\lambda_a = 0$ at $t = 0$. The anticlumping force generated by $\mathcal{P}_h$ is equal to:

$$\frac{\partial \mathcal{P}_h}{\partial \mathbf{x}_a} = -\sum_b m_a m_b c_p^2 \frac{\lambda_a + \lambda_b}{\rho_0^2} h\frac{\partial \nabla W_{ab}}{\partial h} \tag{2.74}$$

The benefit of using anticlumping term over anticlumping force is that the perturbed energy is conserved and in some cases is close to the original energy [5]. Unfortunately, that did not prove to be the case in our simulations.

Another problem that arises in SPH is noise in the pressure field. A common way to address this problem is to use an artificial diffusive term in the SPH continuity equation 2.13. A nice summary and analysis of various difusive terms can be found in Colagrossi et al. [40]. The continuity equation with added diffusive term then becomes:

$$\frac{D\rho_a}{Dt} = -\sum_b m_b \mathbf{v}_{ab} \cdot \nabla W_{ab} + \xi h c \mathcal{D}_a \tag{2.75}$$

where h is smoothing length, c is the numerical sound speed, ξ determines the strength of the diffusion and in our all simulations it is kept equal to $\xi = 0.02$, $\mathcal{D}_a$ is the diffusive term than can vary, we have chosen the term proposed by Colagrossi and Molteni [41]:

$$\mathcal{D}_a = 2\sum_b (\rho_b - \rho_a)\frac{m_b}{\rho_b}\frac{\mathbf{x}_{ab} \cdot \nabla W_{ab}}{|\mathbf{x}_{ab}|^2} \tag{2.76}$$

Adding this term removes the noise and shows good agreement with the analytical solution in the 3.1 Flow in channel example.

3 Numerical results

It is important to state, that the results presented in this chapter are more illustrative and the main goal of this chapter is to illustrate the capabilities of our method. The results are at best qualitative however, with some changes to our implementation and method that will be discussed later, we believe the results could be improved to match the literature and experiments. The list of all parameters used in our simulations can be found in the appendix 1.

3.1 Flow in a channel

This is the simplest case. A parabolic velocity profile is set at the inflow and we are interested if we recover the same velocity profile at the middle of the channel. The velocity profile is given by:

$$v(0, y) = 4.0U_{max}\frac{y(H-y)}{H^2} = U_{max}\frac{4.0}{0.1681}y(0.41-y) \tag{3.1}$$

where U_{max} is the maximum inflow velocity and $H = 0.41$ is the height of the channel. The no-slip boundary condition is set at the walls of the channel. The particles leave the domain at the outflow with no boundary condition set. We will return to this in the next chapter as it might play an important role in achieving better precision.

The time step size is chosen according to the Courant-Friedrichs-Lewy (CFL) condition [42, 43] as $dt = 0.1dr/c$ where $dr = 3.9\text{e-}3$ is the initial distance between the particles and c is the sound speed given by [5]:

$$c = \sqrt{c_0^2 + 4/3c_s^2} \tag{3.2}$$

where $c_0 = 40$ is the numerical sound speed, $c_s = \sqrt{6\nu/\tau}$ [1] is the shear sound speed $\tau = 0.02$ is the relaxation time and $\nu = 0.001$ is the kinematic viscosity of the fluid.

The simulation is carried out for $U_{max} = 1.5$ and we consider the three following cases:

- SHTC-SPH without tensile penalty and artificial diffusion
- SHTC-SPH with tensile penalty and without artificial diffusion
- SHTC-SPH with tensile penalty and artificial diffusion

The resulting velocity profiles are measured in the middle section of the channel and are shown in Figure 3.2. The need for tensile penalty is apparent, as particles start to clump together and void regions start to appear in the first example see Figure 3.1. Therefore tensile penalty is added to prevent the particle clumping. The artificial diffusion - a common practice in SPH [37, 38, 36, 34] - is introduced to stabilize the pressure field, for more detailed information, see [40].

Our results are in good agreement with the analytical solution and the need for tensile penalty and artificial diffusion became apparent. Adding these two terms

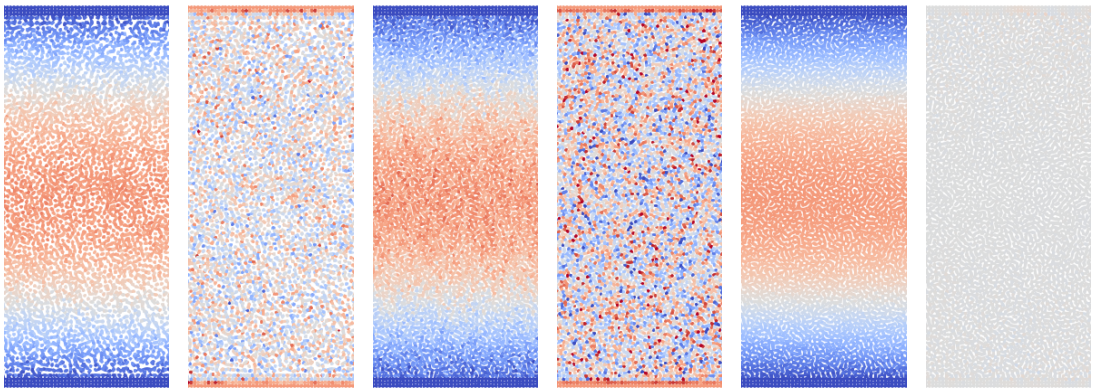


Figure 3.1 The velocity field and the pressure field for the three cases. First two images show the velocity and the pressure field for the case without tensile penalty and artificial diffusion. The second two images show the velocity and the pressure field for the case with tensile penalty and without artificial diffusion. The last two images show the velocity and the pressure field for the case with tensile penalty and artificial diffusion.

to the simulation leads to a more stable simulation that matches the analytical solution.

Even with the enhanced numerical scheme we did not managed to achieve a completely smooth pressure field. One of the reasons might be the fact that we did not prescribe any boundary condition at the outflow effectively leading to the particles being sucked out of the domain. This leads to the formation of pressure waves that propagate through the domain. A possible solution would be to prescribe a zero pressure boundary condition at the outflow. We will return to this in the next chapter.

3.2 Oscillating beam

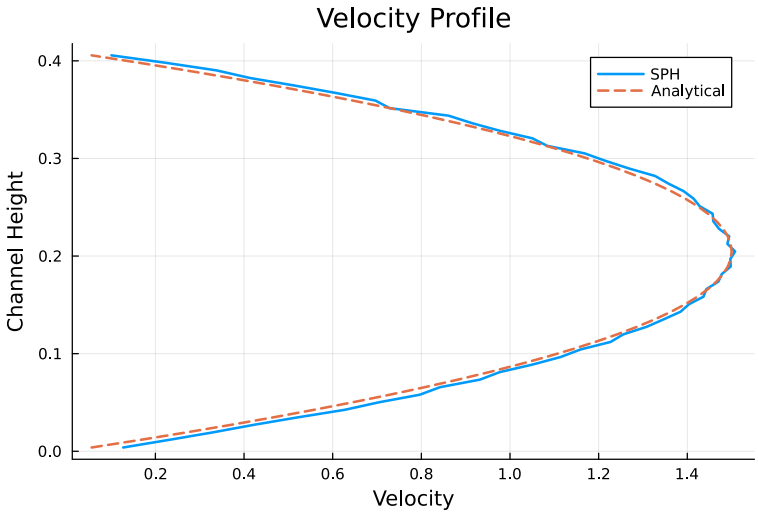
In this section, we have concluded two tests. The first test is an elastic vibration of a beryllium plate [5, 44]. The plate with length $L = 0.06$ and width $W = 0.02$ is defined in the following domain $\Omega = [-0.03, 0.03] \times [-0.005, 0.005]$. It is given the initial velocity of $\mathbf{v}_0(\mathbf{x}) = (0, v_0(x))$ where

$$v_0(x) = A\omega \left[a_1(\sinh(x') + \sin(x')) - a_2(\cosh(x') + \cos(x'))\right] \tag{3.3}$$

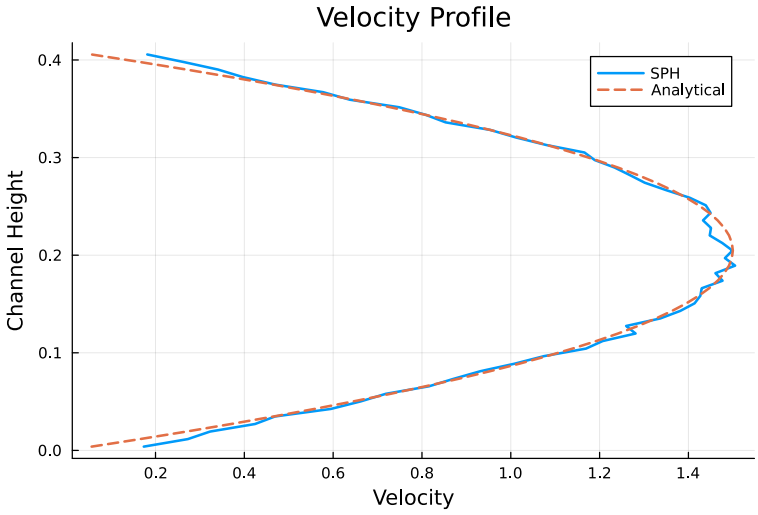
and $x' = \alpha(x + L/2)$, $\alpha = 78.834$, $A = 4.3369\text{e-}5$, $\omega = 2.3597\text{e}5$, $a_1 = 56.6368$ and $a_2 = 57.6455$. The sound speeds are set to $c_s = c_0 = 9046.59$ [5]. The time step is selected according to the CFL condition as $dt = 0.05dr/c$ where $dr = W/40$ and c is the sound speed given by 3.2.

Figure 3.3 shows that energy is conserved and we get a good agreement with the referential solution of [44]. Moreover, the tensile penalty taht is set as $c_p = 4.0c_0$ has a negligible effect on the total energy of the system.

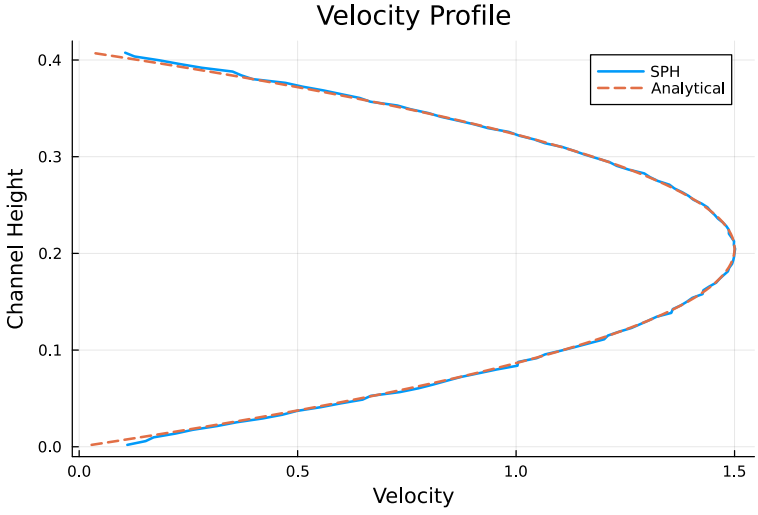
The second test, which is meant to simulate the bending of the elastic beam in the Turek-Hron benchmark [8] acts as a showcase that underlines some of the issues with the SHTC-SPH method. In the test, an elastic beam with length $L = 0.35$



(a) SHTC-SPH without tensile penalty and artificial diffusion. Relative error: 0.0169

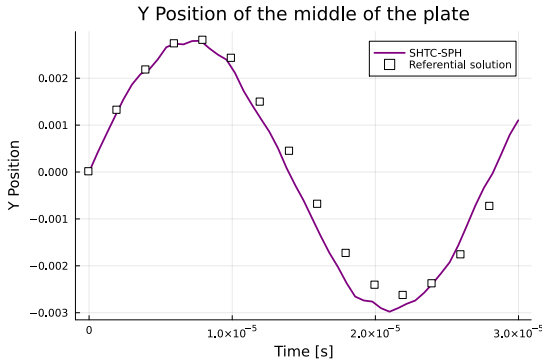


(b) SHTC-SPH with tensile penalty. Relative error: 0.0351

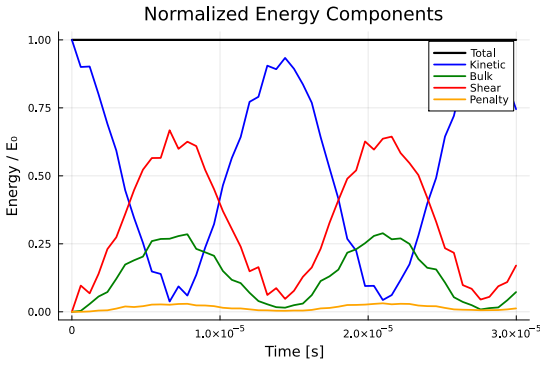


(c) SHTC-SPH with tensile penalty and artificial diffusion. Relative error: 0.0336

Figure 3.2 Comparison of SPH and analytical velocity profiles for parabolical velocity profile at inflow. Note that if we do not use tensile penalty and artificial diffusion, the results are very close to the analytical solution and even better than with tensile penalty however the simulation exhibits unphysical behavior. Relative error is measured in ℓ^2 norm.



(a) Y-position of the middle of the beryllium plate over time.



(b) Energy components of the beryllium plate over time. Tensile penalty plays a negligible role in the overall energy.

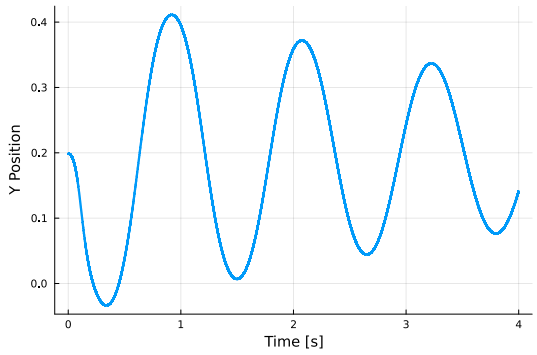
Figure 3.3 The beryllium plate is given initial velocity and then oscillates freely. The number of particles is roughly 10000.

and width $W = 0.02$ is subjected to a gravitational force and then oscillates freely. We are especially interested in whether the SHTC-SPH method is able to simulate large deformations of the elastic beam that occur in the Turek-Hron benchmark. However, this already proves to be a challenge with SHTC-SPH for two main reasons:

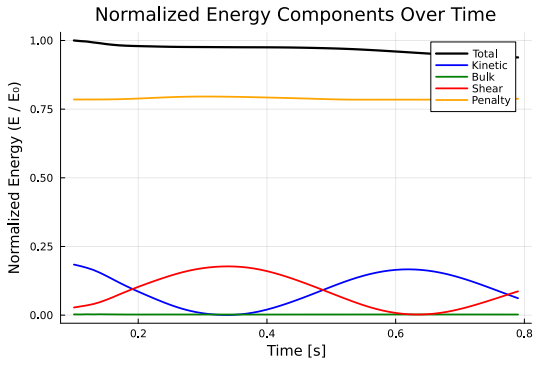
- The sound speed is very high [5] for elastic materials, and the time step is required to be very small, rendering the simulation computationally expensive. This problem could become less significant with better use of parallelization and GPU acceleration, which is currently not implemented in our code and will be one of our future goals.

- Due to the nature of SHTC description of the material, there is no referential configuration which makes the beam prone to breaking when subjected to large deformations. For this reason, when modelling elastic materials, the SPH in Total Lagrangian formalism [45] is preferred. It is of course possible to use the referential configuration in SHTC-SPH to calculate the distortion $\mathbb{A}$ treating it as the inverse of the deformation gradient. This would greatly reduce the breaking and tearing of the beam, however, it defeats the purpose of having one single model for fluids and solids.

The time step is selected according to the CFL condition as $dt = 0.05dr/c$ where $dr = 3.9\text{e-}3$ is the initial distance, $c_s = 40$, $c_0 = 60$ and $c_p = 3.5$. It is important to note two things: the number of particles plays a role in stability. The tearing becomes less significant with more particles, however, this comes at a cost of computational time. The Dirichlet boundary condition is implemented by setting the velocity of the particles at the base of the beam to zero. This might not be the best aproach and a different treatment of the Dirichlet boundary condition might be needed.

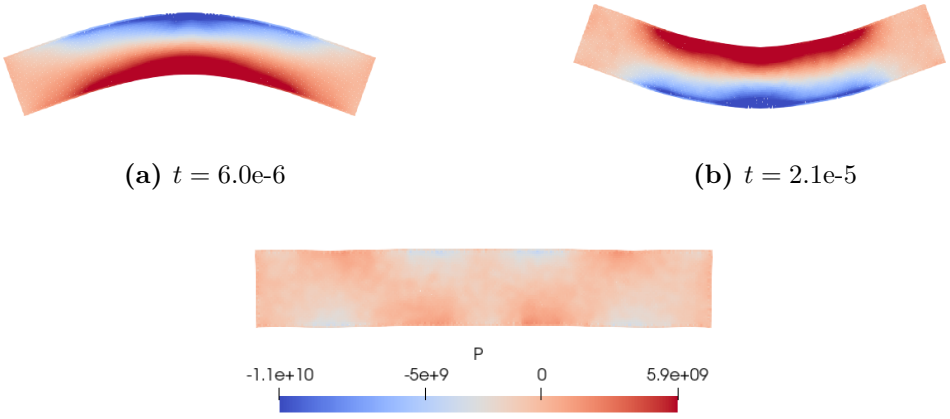


(a) Y-position of the tip of the elastic beam over time.



(b) Energy components of the elastic beam over time. Tensile penalty plays a significant role in the overall energy.

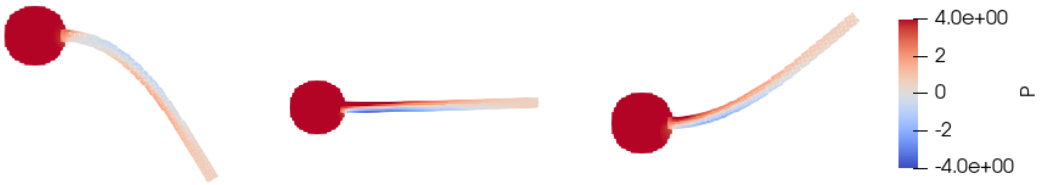
Figure 3.4 The elastic beam is subjected to a gravitational force $g = 10$ for time $t = 0.1$ and then oscillates freely. The number of particles is 460 and the time step is chosen according to CFL condition as $dt = 0.05dr/c$ where $dr = 3.9\text{e-}3$ is the initial distance between the particles and c is the sound speed given by 3.2.



(a) $t = 6.0\text{e-}6$ **(b)** $t = 2.1\text{e-}5$

(c) $t = 2.82\text{e-}5$

Figure 3.5 The pressure field in the beryllium plate.



(a) $t = 0.33$ **(b)** $t = 0.64$ **(c)** $t = 0.79$

Figure 3.6 The elastic beam is subjected to a gravitational force $g = 10$ for time $t = 0.1$ and then oscillates.

3.3 Flow around cylinder

The following benchmark [46] is a simulation of a flow around a circular obstacle in 2D using SHTC-SPH. The simulation parameters are the same as in 3.1 Flow in a channel example. The domain is a rectangular channel without a circular cylinder $\Omega = [0, 2.2] \times [0, 0.41] \setminus B_r(0.2, 0.2)$ with $r = 0.05$. The no-slip boundary condition is set at the walls of the channel and the cylinder. No boundary condition is prescribed at the outflow.

A parabolic velocity profile

$$v(0, y) = \max\{1, t/t_{acc}\} 4.0 U_{max} \frac{y(H - y)}{H^2} \tag{3.4}$$

is set at the inflow, $H = 0.41$ is the height of the pipe. The velocity profile starts at zero and steadily increases until it reaches its maximal value at t_{acc}. We carry out the simulation for Reynolds numbers $Re = 20$ and $Re = 100$. In the first case, we expect to see laminar flow. In the second case, we expect to see time periodic behaviour with the formation of Von Karman vortexes behind the cylinder. The Reynolds number is defined as [46]:

$$Re = 2/3 \frac{U_{max} L}{\nu} \tag{3.5}$$

where U_{max} is the maximum velocity at the inflow, $L = 2r$ is the characteristic length of the flow configuration and ν is the kinematic viscosity of the fluid. For $Re = 20$ example U_{max} is equal to 0.3 and for $Re = 100$ it is equal to 1.5.

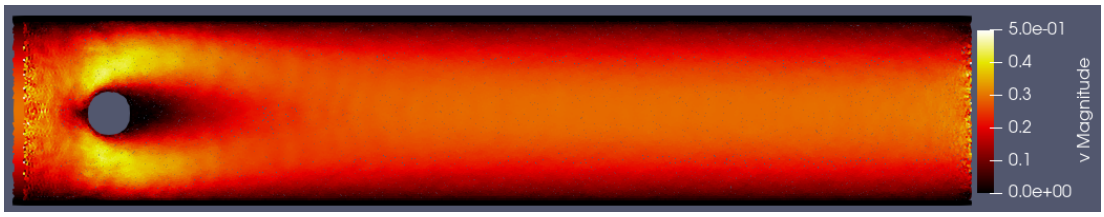


Figure 3.7 Time development of the flow around a cylinder for Re=20 at time t = 30 seconds. No vortexes are present and the flow is laminar. Notice the noise in the velocity field near the inflow and outflow.

The lack of outflow boundary condition became apparent in this case as the fluid particles are sucked out of the domain until the inflow velocity reaches it's maximum. Even then, the pressure field is very noisy, and there is noise in the velocity field near the inflow. This problem vanishes for the case with $Re = 100$.

For the $Re = 20$ case the lift and drag coefficients defined as

$$C_D = \frac{2}{U_{mean}^2 L} F_D \text{ and } C_L = \frac{2}{U_{mean}^2 L} F_L \tag{3.6}$$

where F_D and F_L are the drag and lift forces acting on the cylinder, U_{mean} is the mean velocity at the inflow and L is the characteristic length of the flow configuration were calculated to be $C_D = 7.721$ and $C_L = 0.066$ yielding 38.4% relative error compared to the referential results $C_D = 5.57953523384$ and $C_L = 0.010618948146$ [47].

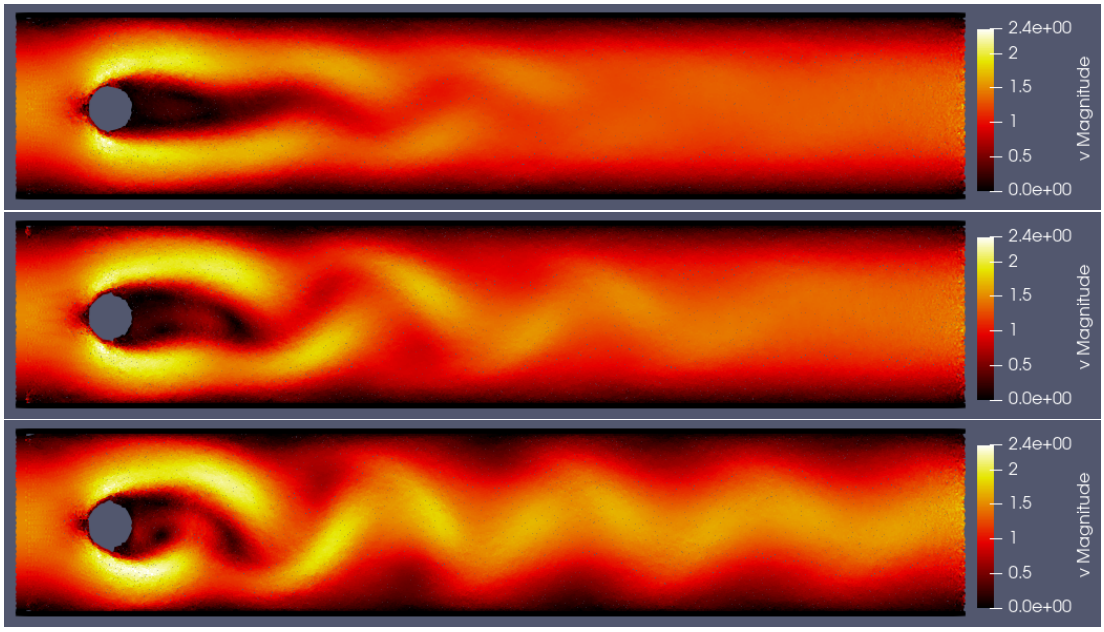


Figure 3.8 Time development of the flow around a cylinder for Re=100 at times t = 18, 20, 30 seconds. Velocity magnitude is shown in the color map and shows the development of Von Karman vortexes.

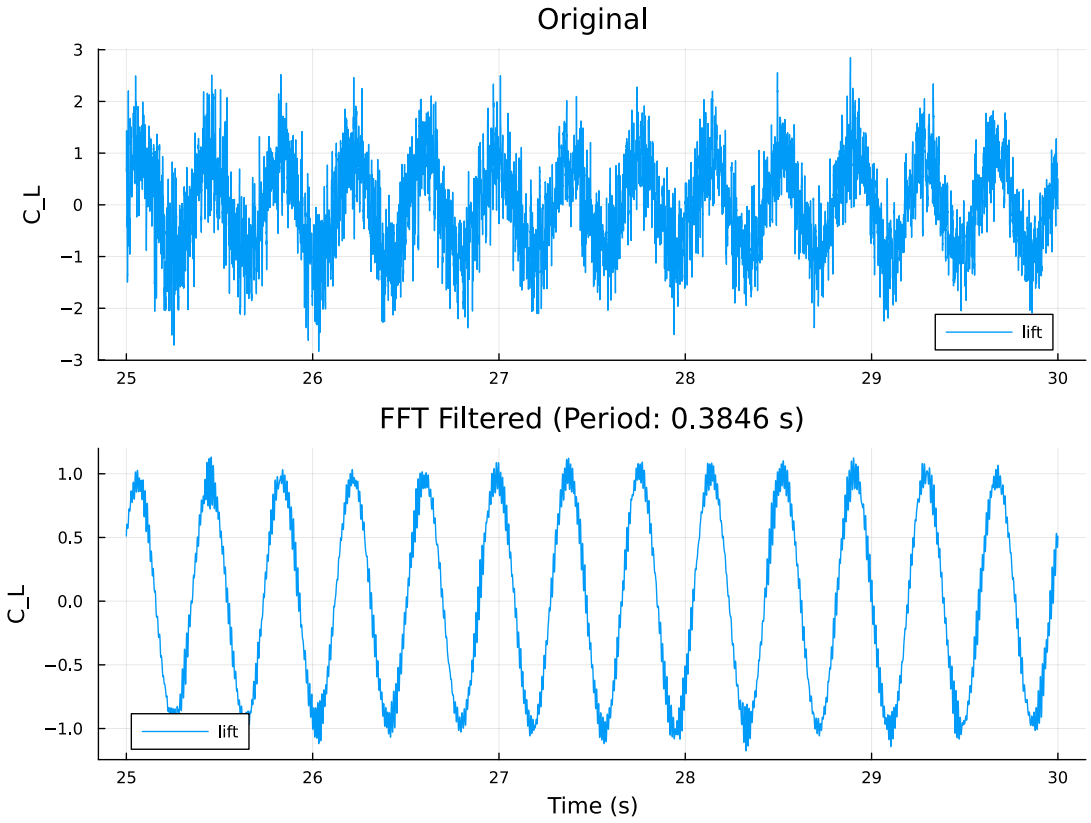


Figure 3.9 The lift coefficient C_L and it's dominant frequency obtained with FFT filter. The detected period was 0.3846 seconds.

For the case with $Re = 100$ vortexes start to form behind the cylinder, and the flow becomes time periodic. We have measured the lift and drag coefficients and the plot of the lift coefficient is shown in Figure 3.9. The noise in the drag coefficient might be due to the fact that at $Re = 100$ the pressure field is very noisy.

3.4 Turek-Hron benchmark

The Turek-Hron benchmark [8] is an extension of the Flow around cylinder benchmark [46] and is perhaps the best known FSI benchmark. The benchmark consists of an elastic beam attached to a cylinder in a rectangular channel with the same dimensions as in the Flow around cylinder benchmark. The main goal of the benchmark is to achieve oscillating motion of the elastic beam driven by the Von Karman vortexes that form behind the cylinder.

The cylinder has a diameter of 0.05 and its center is located at $(0.2, 0.2)$. This means that the cylinder is slightly offset from the center of the channel to prevent dependence of the result on the numerical precision of the simulation.

The elastic beam attached to the cylinder has a length of $L = 0.35$ and a width of $W = 0.02$. The parameters of the simulation are the same as in the Flow around cylinder example and The Oscillating beam example. The complete set of parameters can be found in the Appendix.

Our biggest concern when simulating the Turek-Hron benchmark is the elastic beam. It seems that the beam undergoes irreversible deformations and starts to tear apart Figure 3.10. To rule out the influence of the discretization we have tried to increase the number of particles in the beam. The problem prevailed even with if we made the discretization length three times smaller. We have also run the simulation with different time step sizes and different values of the sounds speeds c_0 and c_s two hundred times of the original values. We have observed that the finer time step size did have only negligible effect on the overall stability of the beam. The increased sound speeds had some effect on the overall stability of the beam, but the problem with particles being torn out of the beam still persisted.

This raises an important question: What is the cause of this and can it be prevented. An obvious solution that, however defeats the purpose of having one single model for fluids and solids is to use distortion as the inverse of the deformation gradient. Another treatment found in Viouleau [17] is artificial surface tension that prevents unnatural cavitation near the surface. This solution would at least partially maintain the elegance of the SHTC model.

It is also possible to make the resolution of the beam finer and use varying time step size for the beam and the fluid [42]. When using this approach the different resolution of the beam and the fluid has to be taken into account by modifying the smoothing lengths accordingly for more details see [43]. This would also have the advantage of achieving better resolution near the surface of the cylinder, that is quite noisy as can be seen in Figure 3.10 and 3.11. These solutions will be the subject of our future work.

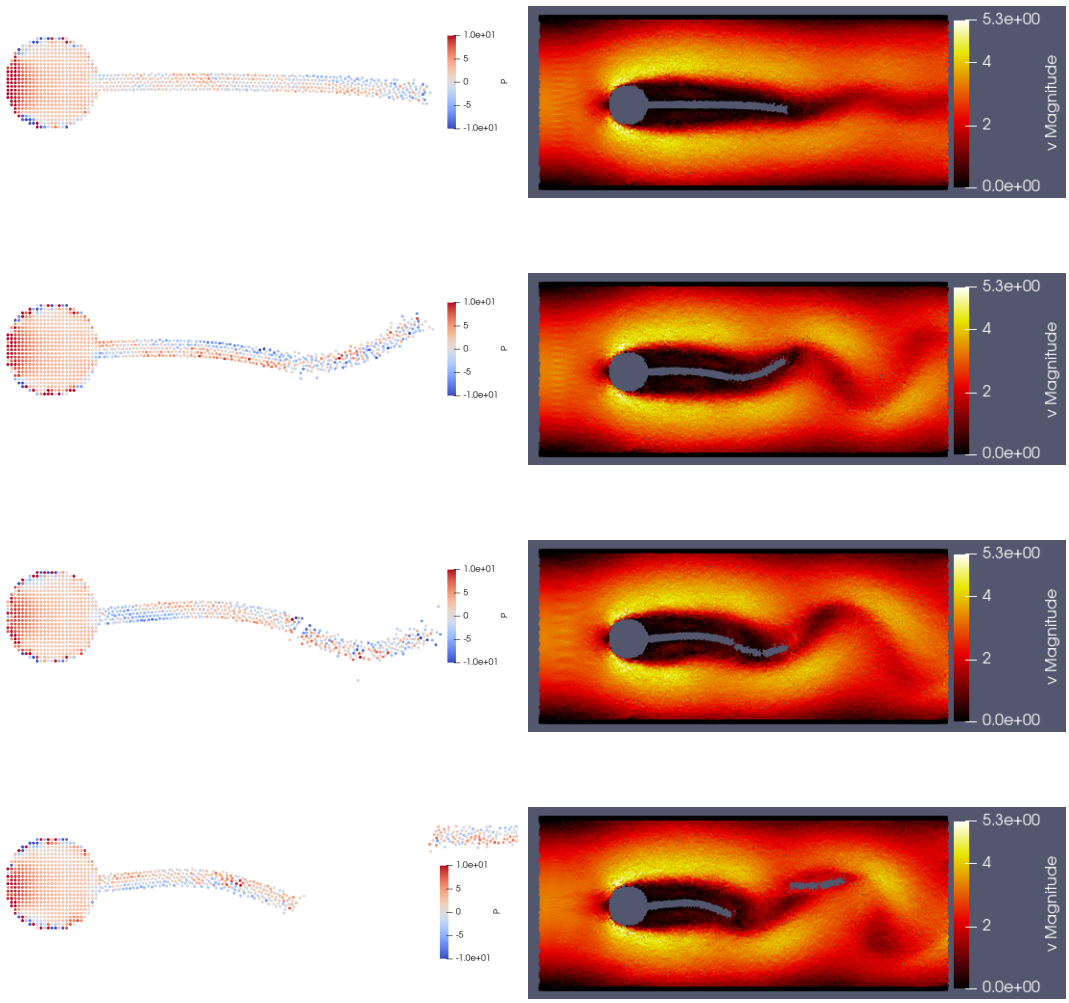
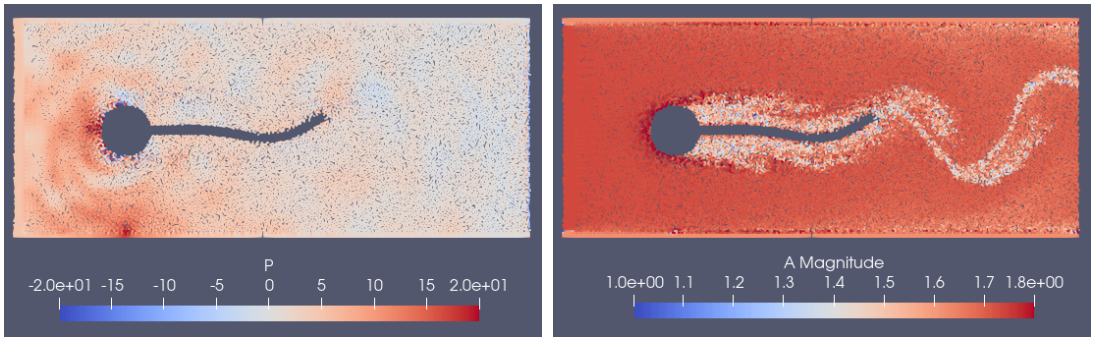


Figure 3.10 Our attempt to simulate the Turek-Hron benchmark. The left column shows the pressure field in the elastic beam and the right column shows the velocity field in the fluid. The simulation parameters are the same as in 3.1 Flow in a channel and 3.2 Oscillating beam examples. The Reynolds number is set to $Re = 220$. Time to accelerate the flow to its full speed is set to $t_{acc} = 0.25$. The images are taken at times $t = 0.55$, $t = 1.35$, $t = 1.55$ and $t = 1.7$ seconds.



(a) The pressure field. **(b)** Frobenius norm of the distortion field.

Figure 3.11 The pressure field is still very noisy especially near the cylinder. The red peak in the pressure field near the lower side of the domain is unphysical and caused by a particle that got stuck near the wall for an unknown reason.

Conclusion & Discussion

In this thesis we have briefly reviewed the SPH method and the SHTC continuum model. We have presented and summarized the SHTC-SPH method [5] and we have put it on test on several benchmark problems - Flow in a channel, Flow around a cylinder and Oscillating elastic beam. We have then attempted to simulate the FSI Turek-Hron benchmark using SHTC-SPH and we have assesed the results of our attempts. Our results are promising and we strongly believe that with some modifications to our method we will be able to simulate the Turek-Hron benchmark successfully in the future.

When attempting to simulate the Turek-Hron benchmark, we have encountered several problems that we have not been able to resolve. These problems fall into two categories:

- Problems that arise due to the SPH method
- Problems that arise due to the SHTC model

The first problem that arises from the use of SPH is the tensile instability. This causes trouble even in the simple case of the flow around a cylinder. Increasing tensile penalty can be used to control this to some extent, however the resulting simulations are not physically correct. Using particle shifting techniques should help with this problem and implementation of this technique is our goal for the future.

The second problem is the irreversible deformation of the elastic beam that occurs when the beam is subjected to large deformations. The deformations seem to be independent of the time step and the number of particles. This problem also occurs if the sound speeds of the fluid and the elastic beam differ more than 200 times. This raises interesting quesion whether this happens due to our implementation, or is it a problem of the SHTC model itself. Simulating elastic materials with SPH is usually done using the referential configuration, and this brings up another question - is it possible to simulate elastic materials with SHTC-SPH?

All test cases presented in [5] are done for relatively small deformations so this remains an open question. One would expect that due to the lack of the referential configuration, any deformation that happens to the elastic beam, however small would be irreversible i.e. the beam would not return to its original shape. This might over time lead to the tearing and breaking of the beam.

Furthermore, we have observed that the particles in the beam try to stick to the tail even though they are already separated and are advected by the fluid. This is clearly a problem that needs to be resolved. This, of course, raises the question whether this phenomenon will play a role in the simulation of more complex materials that behave partly as a fluid and partly as a solid.

In the future, we would like to investigate the cause of this problem further. To differentiate the possible cause, we suggest to run the simulation with distortion being the inverse of the deformation gradient. If the problem persists, it is likely that the problem is in the SHTC model. In the future, we would also like to cross-check the implementation of our SPH library with some alternative software eg. [43] to rule out any implementation errors.

Bibliography

1. Dumbser, Michael; Peshkov, Ilya; Romenski, Evgeniy; Zanotti, Olindo. High order ADER schemes for a unified first order hyperbolic formulation of Newtonian continuum mechanics coupled with electro-dynamics. *Journal of Computational Physics*. 2017, vol. 348, pp. 298–342. ISSN 0021-9991. Available from DOI: https://doi.org/10.1016/j.jcp.2017.07.020.
2. Peshkov, Ilya; Romenski, Evgeniy. *A hyperbolic model for viscous Newtonian flows*. 2014. Available from arXiv: 1403.8068 [physics.flu-dyn].
3. Peshkov, Ilya; Romenski, Evgeniy; Dumbser, Michael. A Unified Hyperbolic Formulation for Viscous Fluids and Elastoplastic Solids. In: 2018, pp. 451–463. ISBN 978-3-319-91547-0. Available from DOI: 10.1007/978-3-319-91548-7_34.
4. Kincl, Ondřej. *Extension of smoothed particle hydrodynamics based on Poisson brackets*. 2024. PhD thesis. Charles University.
5. Kincl, Ondřej; Peshkov, Ilya; Pavelka, Michal; Klika, Václav. Unified description of fluids and solids in Smoothed Particle Hydrodynamics. *Applied Mathematics and Computation*. 2023, vol. 439, p. 127579. ISSN 0096-3003. Available from DOI: https://doi.org/10.1016/j.amc.2022.127579.
6. Kincl, Ondřej; Pavelka, Michal. Globally time-reversible fluid simulations with smoothed particle hydrodynamics. *Computer Physics Communications*. 2023, vol. 284, p. 108593. ISSN 0010-4655. Available from DOI: https://doi.org/10.1016/j.cpc.2022.108593.
7. Kincl, Ondřej; Schmoranzer, David; Pavelka, Michal. Simulation of superfluid fountain effect using smoothed particle hydrodynamics. *Physics of Fluids*. 2023, vol. 35. Available from DOI: 10.1063/5.0145864.
8. Turek, Stefan; Hron, Jaroslav. Proposal for Numerical Benchmarking of Fluid–Structure Interaction Between an Elastic Object and Laminar Incompressible Flow. In: 2007, vol. 53, pp. 371–385. ISBN 978-3-540-34595-4. Available from DOI: 10.1007/3-540-34596-5_15.
9. Peshkov, Ilya; Dumbser, Michael; Boscheri, Walter; Romenski, Evgeniy; Chiocchetti, Simone; Ioriatti, Matteo. Simulation of non-Newtonian viscoplastic flows with a unified first order hyperbolic model and a structure-preserving semi-implicit scheme. *Computers & Fluids*. 2021, vol. 224, p. 104963. ISSN 0045-7930. Available from DOI: https://doi.org/10.1016/j.compfluid.2021.104963.
10. Peshkov, Ilya; Boscheri, Walter; Loubère, Raphaël; Romenski, Evgeniy; Dumbser, Michael. Theoretical and numerical comparison of hyperelastic and hypoelastic formulations for Eulerian non-linear elastoplasticity. *Journal of Computational Physics*. 2019, vol. 387, pp. 481–521. ISSN 0021-9991. Available from DOI: https://doi.org/10.1016/j.jcp.2019.02.039.
11. Godunov, S.K.; Romenskii, E.I. Nonstationary equations of nonlinear elasticity theory in eulerian coordinates. *Journal of Applied Mechanics and Technical Physics*. 1972, vol. 13, pp. 868–884. Available from DOI: https://doi.org/10.1007/BF01200547.

12. Besseling, J. F. A Thermodynamic Approach to Rheology. In: Parkus, H.; Sedov, L. I. (eds.). *Irreversible Aspects of Continuum Mechanics and Transfer of Physical Characteristics in Moving Fluids*. Vienna: Springer Vienna, 1968, pp. 16–53. isbn 978-3-7091-5581-3.

13. Godunov, Sergei; Romenski, Evgeniy. *Elements of Continuum Mechanics and Conservation Laws*. 2003. isbn 978-1-4419-3399-7. Available from doi: 10.1007/978-1-4757-5117-8.

14. Frenkel, D. *Kinetic Theory Of Liquids*. 1946.

15. Monaghan, J.J; Gingold, R.A. Shock simulation by the particle method SPH. *Journal of Computational Physics*. 1983, vol. 52, no. 2, pp. 374–389. issn 0021-9991. Available from doi: https://doi.org/10.1016/0021-9991(83)90036-0.

16. Holger., Wendland. *Piecewise polynomial, positive definite and compactly supported radial functions of minimal degree*. [N.d.]. Available also from: http://gen.lib.rus.ec/book/index.php?md5=84e6dfd97d4677353a8ddc0b5b30653c

17. Violeau, Damien. *Fluid Mechanics and the SPH Method: Theory and Applications*. Oxford University Press, 2012. isbn 9780199655526. Available from doi: 10.1093/acprof:oso/9780199655526.001.0001.

18. Monaghan, J. J. Smoothed Particle Hydrodynamics. *Annual Review of Astronomy and Astrophysics*. 1992, vol. 30, no. Volume 30, 1992, pp. 543–574. issn 1545-4282. Available from doi: https://doi.org/10.1146/annurev.aa.30.090192.002551.

19. Colagrossi, Andrea; Durante, Danilo; Bonet Avalos, Josep; Souto-Iglesias, Antonio. Discussion of Stokes' hypothesis through the smoothed particle hydrodynamics model. *Phys. Rev. E*. 2017, vol. 96, p. 023101. Available from doi: 10.1103/PhysRevE.96.023101.

20. Morris, Joseph P.; Fox, Patrick J.; Zhu, Yi. Modeling Low Reynolds Number Incompressible Flows Using SPH. *Journal of Computational Physics*. 1997, vol. 136, no. 1, pp. 214–226. issn 0021-9991. Available from doi: https://doi.org/10.1006/jcph.1997.5776.

21. Wu, Ke; Yang, Dongmin; Wright, Nigel. A coupled SPH-DEM model for fluid-structure interaction problems with free-surface flow and structural failure. *Computers & Structures*. 2016, vol. 177, pp. 141–161. issn 0045-7949. Available from doi: https://doi.org/10.1016/j.compstruc.2016.08.012.

22. Chen, Cheng; Shi, Wen-Kui; Shen, Yan-Ming; Chen, Jian-Qiang; Zhang, A-Man. A multi-resolution SPH-FEM method for fluid-structure interactions. *Computer Methods in Applied Mechanics and Engineering*. 2022, vol. 401, p. 115659. issn 0045-7825. Available from doi: https://doi.org/10.1016/j.cma.2022.115659.

23. Xu, Yixiang; Yang, Gang; Zhu, Yawei; Hu, Dean. A coupled SPH-FVM method for simulating incompressible interfacial flows with large density difference. *Engineering Analysis with Boundary Elements*. 2021, vol. 128, pp. 227–243. issn 0955-7997. Available from doi: https://doi.org/10.1016/j.enganabound.2021.04.005.

24. Antoci, Carla; Gallati, Mario; Sibilla, Stefano. Numerical simulation of fluid–structure interaction by SPH. *Computers & Structures*. 2007, vol. 85, no. 11, pp. 879–890. ISSN 0045-7949. Available from DOI: https://doi.org/10.1016/j.compstruc.2007.01.002. Fourth MIT Conference on Computational Fluid and Solid Mechanics.

25. Adami, S.; Hu, X.Y.; Adams, N.A. A generalized wall boundary condition for smoothed particle hydrodynamics. *Journal of Computational Physics*. 2012, vol. 231, no. 21, pp. 7057–7075. ISSN 0021-9991. Available from DOI: https://doi.org/10.1016/j.jcp.2012.05.005.

26. Zhang, Chi; Rezavand, Massoud; Hu, Xiangyu. A multi-resolution SPH method for fluid-structure interactions. *Journal of Computational Physics*. 2021, vol. 429, p. 110028. ISSN 0021-9991. Available from DOI: https://doi.org/10.1016/j.jcp.2020.110028.

27. Han, Luhui; Hu, Xiangyu. SPH modeling of fluid-structure interaction. *Journal of Hydrodynamics*. 2018, vol. 30, pp. 62–69. Available from DOI: 10.1007/s42241-018-0006-9.

28. Pavelka, Michal; Klika, Václav; Grmela, Miroslav. *Multiscale Thermo-Dynamics: Introduction to GENERIC*. 2018. ISBN 9783110350951. Available from DOI: 10.1515/9783110350951.

29. Pavelka, Michal; Klika, Václav; Grmela, Miroslav. Time reversal in nonequilibrium thermodynamics. *Phys. Rev. E*. 2014, vol. 90, p. 062131. Available from DOI: 10.1103/PhysRevE.90.062131.

30. E., Hairer. *Numerical geometric integration of ODE*. Web draft. 1999. Available also from: http://gen.lib.rus.ec/book/index.php?md5=2c68de91f5f7e0559afbae9715aa6e08.

31. Swegle, J.W.; Hicks, D.L.; Attaway, S.W. Smoothed Particle Hydrodynamics Stability Analysis. *Journal of Computational Physics*. 1995, vol. 116, no. 1, pp. 123–134. ISSN 0021-9991. Available from DOI: https://doi.org/10.1006/jcph.1995.1010.

32. Monaghan, J.J. SPH without a Tensile Instability. *Journal of Computational Physics*. 2000, vol. 159, no. 2, pp. 290–311. ISSN 0021-9991. Available from DOI: https://doi.org/10.1006/jcph.2000.6439.

33. Lahiri, Saptarshi Kumar; Bhattacharya, Kanishka; Shaw, Amit; Ramachandra, L.S. A stable SPH with adaptive B-spline kernel. *Journal of Computational Physics*. 2020, vol. 422, p. 109761. ISSN 0021-9991. Available from DOI: 10.1016/j.jcp.2020.109761.

34. Adami, S.; Hu, X.Y.; Adams, N.A. A transport-velocity formulation for smoothed particle hydrodynamics. *Journal of Computational Physics*. 2013, vol. 241, pp. 292–307. ISSN 0021-9991. Available from DOI: https://doi.org/10.1016/j.jcp.2013.01.043.

35. Wang, Zhentong; Haidn, Oskar J.; Hu, Xiangyu. *The efficient implementation of transport velocity formulation*. 2024. Available from arXiv: 2411.13992 [physics.flu-dyn].

36. Sun, P.N.; Colagrossi, A.; Marrone, S.; Antuono, M.; Zhang, A.M. Multi-resolution Delta-plus-SPH with tensile instability control: Towards high Reynolds number flows. *Computer Physics Communications*. 2018, vol. 224, pp. 63–80. ISSN 0010-4655. Available from DOI: https://doi.org/10.1016/j.cpc.2017.11.016.

37. Marrone, S.; Antuono, M.; Colagrossi, A.; Colicchio, G.; Le Touzé, D.; Graziani, G. delta-SPH model for simulating violent impact flows. *Computer Methods in Applied Mechanics and Engineering*. 2011, vol. 200, no. 13, pp. 1526–1542. ISSN 0045-7825. Available from DOI: https://doi.org/10.1016/j.cma.2010.12.016.

38. Sun, P.N.; Colagrossi, A.; Marrone, S.; Antuono, M.; Zhang, A.-M. A consistent approach to particle shifting in the delta-Plus-SPH model. *Computer Methods in Applied Mechanics and Engineering*. 2019, vol. 348, pp. 912–934. ISSN 0045-7825. Available from DOI: https://doi.org/10.1016/j.cma.2019.01.045.

39. Liu, Wen-Tao; Sun, Pengnan; Ming, Furen; Zhang, A-Man. Application of particle splitting method for both hydrostatic and hydrodynamic cases in SPH. *Acta Mechanica Sinica*. 2018, vol. 34. Available from DOI: 10.1007/s10409-017-0739-7.

40. Antuono, M.; Colagrossi, A.; Marrone, S. Numerical diffusive terms in weakly-compressible SPH schemes. *Computer Physics Communications*. 2012, vol. 183, no. 12, pp. 2570–2580. ISSN 0010-4655. Available from DOI: https://doi.org/10.1016/j.cpc.2012.07.006.

41. Molteni, Diego; Colagrossi, Andrea. A simple procedure to improve the pressure evaluation in hydrodynamic context using the SPH. *Computer Physics Communications*. 2009, vol. 180, no. 6, pp. 861–872. ISSN 0010-4655. Available from DOI: https://doi.org/10.1016/j.cpc.2008.12.004.

42. Zhang, Chi; Rezavand, Massoud; Hu, Xiangyu. Dual-criteria time stepping for weakly compressible smoothed particle hydrodynamics. *Journal of Computational Physics*. 2020, vol. 404, p. 109135. ISSN 0021-9991. Available from DOI: https://doi.org/10.1016/j.jcp.2019.109135.

43. Zhang, Chi; Rezavand, Massoud; Zhu, Yujie; Yu, Yongchuan; Wu, Dong; Zhang, Wenbin; Wang, Jianhang; Hu, Xiangyu. SPHinXsys: An open-source multi-physics and multi-resolution library based on smoothed particle hydrodynamics. *Computer Physics Communications*. 2021, vol. 267, p. 108066. ISSN 0010-4655. Available from DOI: 10.1016/j.cpc.2021.108066.

44. Boscheri, Walter; Chiocchetti, Simone; Peshkov, Ilya. A cell-centered implicit-explicit Lagrangian scheme for a unified model of nonlinear continuum mechanics on unstructured meshes. *Journal of Computational Physics*. 2022, vol. 451, p. 110852. ISSN 0021-9991. Available from DOI: https://doi.org/10.1016/j.jcp.2021.110852.

45. Vignjevic, Rade; Reveles, Juan; Campbell, James. SPH in a Total Lagrangian formalism. *CMES - Computer Modeling in Engineering and Sciences*. 2006, vol. 14, pp. 181–198.

46. Schäfer, M.; Turek, S.; Durst, F.; Krause, E.; Rannacher, R. Benchmark Computations of Laminar Flow Around a Cylinder. In: *Flow Simulation with High-Performance Computers II: DFG Priority Research Programme Results 1993–1995*. Ed. by Hirschel, Ernst Heinrich. Wiesbaden: Vieweg+Teubner Verlag, 1996, pp. 547–566. isbn 978-3-322-89849-4. Available from doi: 10.1007/978-3-322-89849-4_39.

47. Nabh, Guido. *On High Order Methods for the Stationary Incompressible Navier-Stokes Equations*. IWR, 1998.

Parameter	Flow in a channel	Beryllium plate	Oscillating beam	Flow around a cylinder	Turek-Hron benchmark
dr	3.9e-3	5.0e-4	3.9e-3	3.9e-3	3.9e-3
dt	9.74e-6	9.04e-10	5.15e-6	5.15e-6	5.15e-6
c_0	40	9046.59	60	40	40 and 60
c_s	5.47e-3	9046.59	40	5.47e-3	5.47e-3 and 40
ρ	1.0	1.0	1845.0	1.0	1.0 and 1.0
c_p	3.5	36186.36	3.5	3.5	3.5 and 3.5
h	9.75e-3	1.5e-3	9.75e-3	9.75e-3	9.75e-3

Table 1 Simulation parameters for different benchmark cases. The Turek-Hron benchmark uses two values: the first for the fluid and the second for the elastic beam. The artificial diffusion parameter $\xi = 0.02$ is used in all simulations. Reynolds number $Re = 20$ corresponds to $U_{\max} = 0.3$, $Re = 100$ to $U_{\max} = 1.5$, and $Re = 220$ to $U_{\max} = 3.3$ in all examples.